Anchored Causal Action-Contrast Transfer for Cross-Network Traffic Signal Control

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Abstract

Traffic-signal transfer requires reliable action ranking as topology, demand and signal programs change across networks. We introduce anchored Causal-Factored Cross-City Model Transfer (CFCMT), which predicts normalized cost differences between feasible phases from matched simulator states. A parent-restricted rigid anchor and an antisymmetric all-pair model are blended through seed-blocked target validation before deployment. Target counterfactual labels make this an offline-adaptation method. Development on 18 Simulation of Urban MObility (SUMO) networks from seven city groups reduced leave-one-city-out normalized action regret by 9.84% on a held-out seed; six of seven groups improved. On two external networks, an initial 56-rollout matrix showed a 6.27% reduction in all-departed waiting relative to a same-information dense residual baseline. With both models frozen, 80 further 3,600-s rollouts on ten new paired seeds confirmed a 4.93% reduction (paired 95% interval, 2.64–7.31%); nine of ten seed-level contrasts improved. A separate preregistered 64-seed comparison found a 3.04% Jinan gain but a 7.87% Los Angeles regression, rejecting joint two-city benefit over a legacy near-target-only controller. A subsequent target-offline selector returned target-only control for Los Angeles and admitted a Hangzhou source component for Jinan. On 56 further Jinan seeds, the admitted controller reduced waiting by 4.53% relative to a lower-capacity zero-source-row target model. The paired 95% interval was 4.28–4.79%, and all 56 seeds improved. Pressure policies remained stronger. These results support structured action-contrast modelling over dense residual transfer and a reproducible Jinan controller-pair gain, while leaving source-row attribution and broader cross-city benefit unresolved.

1 Introduction

Traffic-signal controllers are commonly trained and tested within one fixed network. New random seeds change arrivals while leaving topology, routes, phase programs and feature semantics largely unchanged. Cross-network deployment changes these conditions together. A model can therefore fit source transitions yet misrank feasible actions in a target network. Action ordering, rather than transition accuracy alone, determines which control decision the model recommends.

Two established approaches expose the challenge. MaxPressure uses local queue imbalance and has throughput and stability guarantees under its stated store-and-forward assumptions (Varaiya, 2013). The PhasePressure comparator likewise uses local queue imbalance and transfers without a city-indexed policy. Pressure structure also underlies controllers learned by reinforcement learning (RL), such as PressLight and MPLight (Wei et al., 2019a; Chen et al., 2020). The Flipping- and Rotation-invariant All-Phase design (FRAP) represents competition between conflicting phases from movement demand (Zheng et al., 2019). These baselines are strong because their inputs directly reflect the queues affected by signal decisions.

Dynamics-Aware Hybrid Offline-and-Online Reinforcement Learning (H2O) combines fixed real-world data with interaction in an imperfect simulator (Niu et al., 2022). Its extension, An

Improved Framework for Hybrid Offline-and-Online Reinforcement Learning with Dynamics Gaps ($H2O^+$), explicitly accounts for dynamics gaps (Niu et al., 2023). Their real–simulator setting motivates our dense residual comparator, although cross-network transfer introduces a different shift. An unrestricted residual can absorb network identity, demand scale and phase-specific associations that change between cities.

Transfer-oriented signal control spans different target information budgets. MetaLight adapts a meta-learned initialization with target experience (Zang et al., 2020). GEneral Scenario-Agnostic reinforcement learning (GESA) normalizes heterogeneous intersections into unified state and action structures (Jiang et al., 2024a). X-Light learns cross-city decision-trajectory representations (Jiang et al., 2024b), whereas CrossLight combines source-city offline pretraining with target-city online fine-tuning (Sun et al., 2024). Sim-to-real work such as Uncertainty-Aware Grounded Action Transformation (UGAT) transforms actions to compensate for a simulator dynamics gap (Da et al., 2023). These methods instantiate distinct approaches to topology normalization, meta-adaptation and dynamics-gap correction. Their zero-shot, online-interaction and sim-to-real protocols answer different questions. Our final comparison uses target offline adaptation, with counterfactual simulator labels collected before deployment and no target closed-loop reward used for model selection.

Causal representation work motivates modular mechanisms whose changes can remain sparse across environments (Bengio et al., 2019; Schölkopf et al., 2021). Factored Adaptation for Non-Stationary Reinforcement Learning operationalizes this idea through a factored Markov decision process and individual latent change factors (Feng et al., 2022). Our earlier mechanism-factored world model restricted each predicted component to declared parents and adapted its mechanism state separately. Traffic control places a further requirement on this structure. Accurate decomposition of absolute dynamics does not guarantee accurate ordering of feasible phases. The final method retains parent restrictions but changes the estimand from next-state prediction to within-state action contrast.

We formulate Causal-Factored Cross-City Model Transfer (CFCMT) around matched action contrasts. At each simulator snapshot, every feasible phase is executed from the same restored state. Candidate costs are measured relative to a source-selected pressure reference and normalized by the matched group's action range. A parent-restricted rigid anchor and an antisymmetric all-pair model estimate these contrasts. Their anchored interpolation (Equation (9)) is selected by out-of-fold (OOF) validation that holds out complete target adaptation seeds. A local confidence cap limits the correction when the models disagree. Definitions, formal statements and inline proofs appear in Sections 4.2, 4.4 and 4.5.

The target information budget defines this adaptation setting. CFCMT uses target-network files, observed target state and counterfactual labels from an uncalibrated Simulation of Urban MObility (SUMO) model. The simulator is not fitted to field trajectories, but its counterfactual labels are target data. The method is therefore calibration-light rather than zero-shot. Adaptation, offline confirmation and closed-loop seeds are disjoint. The evaluation cache was generated after the target model and selector were frozen.

Development uses 18 source networks from seven city groups, followed by evaluation on two external city-derived networks, Los Angeles and Jinan. In 56 matched 3,600-s rollouts, CFCMT is compared with fixed time, simulator-only model predictive control (MPC), a rigid anchor, a dense residual and two pressure policies. All policies share target networks, feasible phases, demand seeds, transition execution and metric populations. An independent confirmation then pairs the unchanged CFCMT and dense models on ten new seeds across the same four scenarios, adding 80 rollouts without refitting or parameter selection. Further controller comparisons distinguish source-plus-target control from target-only control under fixed target adaptation budgets. V91 evaluates the original controller on 64 fresh seeds, four target scenarios and 512 rollouts. V98 independently

confirms a subsequent target-offline selector on 56 further Jinan seeds. Its topology-repaired refit and shared pressure guard distinguish this procedure from the frozen V43 controller. V-prefixed identifiers denote frozen experimental protocols rather than method names.

The Reinforcement Signal Control (RESCO) benchmark (Ault and Sharon, 2021) and LibSignal (Mei et al., 2022) RL runs provide native-protocol integration checks. Their state, reward, action timing and training budgets differ from the paired experiment. Same-protocol reproductions of CrossLight, X-Light and GESA remain a comparison gap.

This study makes four contributions. First, it formulates cross-network signal adaptation around matched action contrasts and proves cancellation of action-common domain bias (Proposition 1). Second, it combines a parent-restricted anchor with an antisymmetric all-action correction and derives exact ranking and bounded perturbation properties (Propositions 2 and 3 and corollary 1). Third, it implements seed-blocked selection and full refitting, with OOF and evaluation models excluded from deployment. Fourth, it separates structured-model gain, controller-pair performance and source-row attribution through distinct frozen comparisons. The central evidence concerns matched action contrasts relative to a same-information dense residual and a separately confirmed Jinan controller-pair gain. Opposite V91 outcomes across the two cities establish the need to retain network-dependent failures. The V98 comparator's lower capacity limits source-row attribution, and PhasePressure remains the stronger operational reference.

2 Results

2.1 A frozen action-contrast transfer protocol

The final experiment evaluated target offline adaptation using matched counterfactuals from restored SUMO states (Figure 1). Target roles were separated by seed, with seeds 5057 and 6067 reserved for adaptation. Seed 8171 was collected after the joint method freeze for one offline action-regret confirmation. Seeds 8081 and 9091 were reserved for closed-loop policy rollouts. OOF models selected the blend; deployment used one refit per external city on all valid adaptation groups. Frozen artifact identities and the regeneration contract are documented in Appendix A.13 and table 18.

The source bank comprised 18 networks in seven city groups: Atlanta, Cologne, Hangzhou, Ingolstadt, New York, two RESCO synthetic grids and Salt Lake City. Together, these networks contained 317 controlled traffic lights and 6,076 non-internal lanes. The external set comprised a four-signal Los Angeles corridor and a 12-signal Jinan grid with three demand files (Figure 2; detailed counts in Appendix A.1 and table 8). Development folds held out entire city groups to prevent related subnetworks from entering training. This excluded every Cologne, Hangzhou or Ingolstadt subnetwork whenever its city was held out.

2.2 Seed-held-out development selected an action-contrast correction

Models and the city-level selector were fitted on source seeds 2027 and 3037, then evaluated on excluded seed 4047. In leave-one-city-out evaluation, city-macro normalized action regret fell from 0.2414 for the rigid anchor to 0.2177 for the selected anchored model. This represented a 9.84% reduction across the seven held-out city groups (Table 1). Six groups improved; Hangzhou regressed by 0.00388, below the frozen 0.01 tolerance.

Earlier variants failed their gates, which tested worst-city performance and city-level stability. A global pairwise correction regressed on the added Salt Lake domain. The first anchored selector exceeded its worst-city tolerance, and a cross-fitted selector failed the city-level stability requirement.

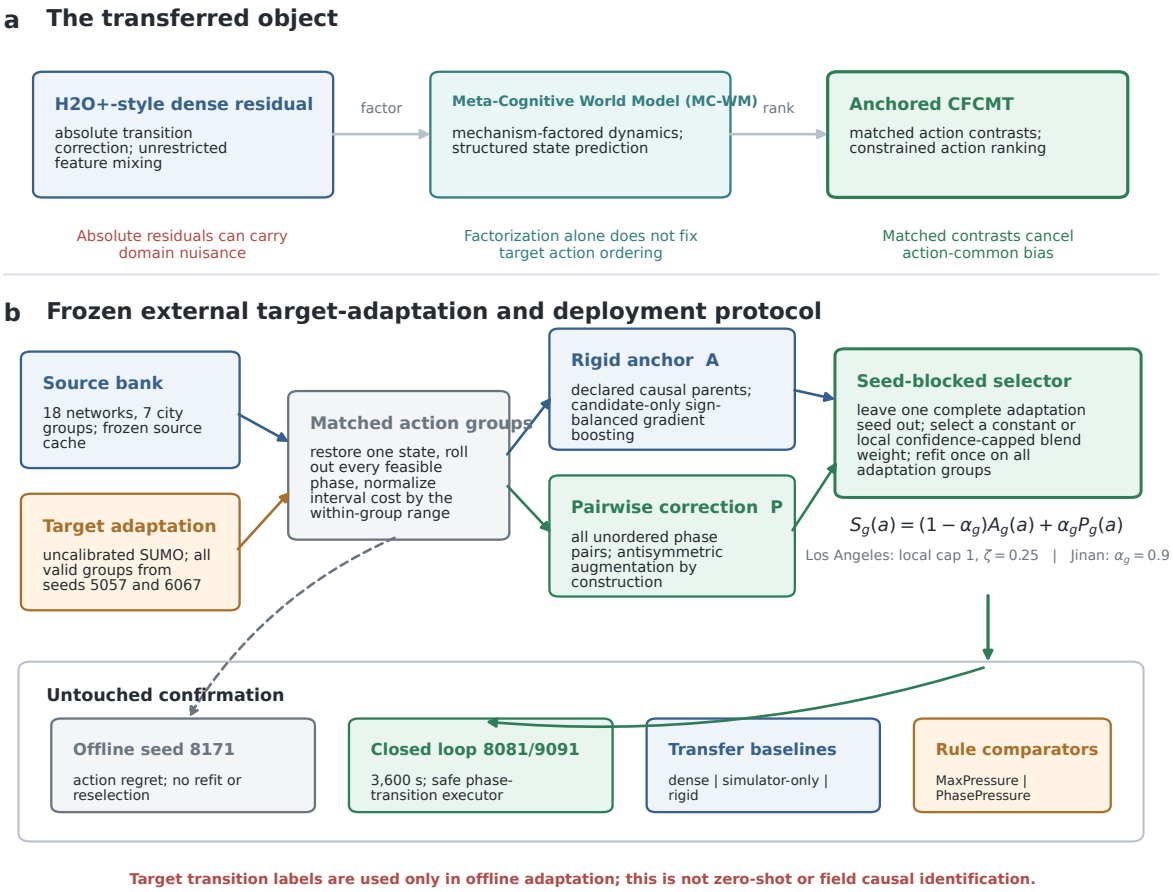


Figure 1: Transferred object and frozen evaluation protocol. **a**, The method lineage compares absolute dense-residual prediction and factored Meta-Cognitive World Model (MC-WM) dynamics with the final matched-ranking estimand. The dense baseline in panel **b** uses the same matched contrast data and target refit budget as CFCMT. **b**, A seed-blocked target selector combines a rigid parent-restricted anchor with an antisymmetric all-pair correction. The pressure reference defines the contrast origin. MaxPressure and PhasePressure remain independent comparators, without serving as fallback actions. Target counterfactual labels make this a target offline adaptation setting.

Table 1: Seed-held-out development confirmation. Regret is candidate cost minus the matched-group optimum, normalized by the within-group action range; lower is better.

Development result	Anchor regret	Selected regret	Relative gain
Seven-city leave-one-city-out	0.241	0.218	9.84%

These outcomes were retained, and the successful seed-held-out protocol was frozen before external labels were generated (Appendix A.2 and table 9).

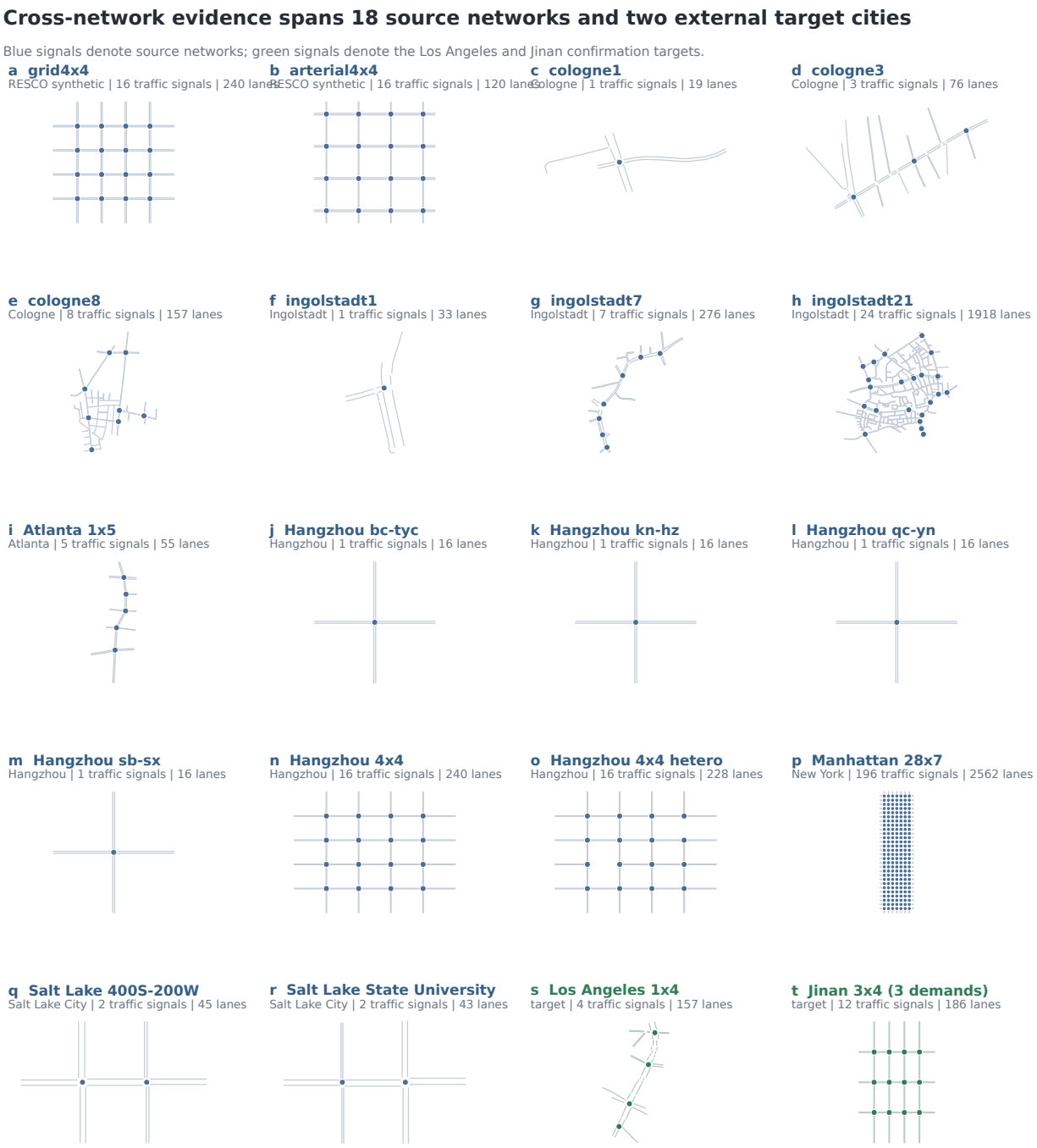


Figure 2: Network shift in development and external confirmation. Lines show non-internal SUMO lanes, and points identify the controlled traffic lights. Blue panels show 18 source networks grouped into seven development cities; green panels show the external target geometries. The Jinan geometry is paired with three demand files for evaluation. These converted networks document topology and action-set variation without representing calibrated city-wide digital twins.

Table 2: External seed-8171 offline confirmation after city-specific method freeze. The legacy near-target-only arm shares the causal action-ranking family and target groups. Its weighted mean has zero deployed source weight, but exact ties retain a source prior. Adaptation counterfactuals come from seeds 5057 and 6067; held-out groups are excluded from fitting and selection. Table gains compare CFCMT with the arm labelled target only.

City	Adaptation groups	Held-out groups	Anchor	Target only	CFCMT	CFCMT vs target only	Selected blend
Los Angeles	165	79	0.148	0.217	0.145	33.24%	local cap 1, $\zeta = 0.25$
Jinan	911	457	0.436	0.404	0.392	2.85%	constant $\alpha_g = 0.9$
External macro	–	2 cities	0.292	0.310	0.269	13.48%	city-specific

2.3 External offline scores favoured source-plus-target fitting

The external selector used all valid adaptation groups: 165 in Los Angeles and 911 across the three Jinan demands. Holding out complete adaptation seeds produced two selection folds for each city. Los Angeles selected a local blend with disagreement cap 1 and confidence multiplier 0.25. Jinan selected a constant correction weight of 0.9 across its demand files. Each deployment artifact was refitted once on all adaptation groups and frozen before seed 8171 was collected.

On seed 8171, Los Angeles regret decreased from 0.1482 to 0.1450 (2.18%), while optimal-action accuracy increased from 0.785 to 0.810. Jinan regret decreased from 0.4361 to 0.3921 (10.10%), while accuracy increased from 0.296 to 0.342 (Table 2; panel **a** of Figure 3). With equal city weights, regret decreased from 0.2922 to 0.2685 (8.09%), with both city point estimates improving.

The freeze included a legacy near-target-only comparator from the same causal action-advantage family. It shared the feature semantics and target adaptation groups, with a target prediction weight numerically equal to one. Its deployed weighted mean excluded source-labelled groups, although a retained source prior could resolve exact action ties. On seed 8171, this comparator achieved regret 0.2172 in Los Angeles and 0.4036 in Jinan. CFCMT reduced the equal-city point estimate from 0.3104 to 0.2685 (13.48%).

The paired action-group improvement over this comparator was 0.07219 in Los Angeles (95% interval $[0.01671, 0.13512]$; two-sided bootstrap probability value 0.011). Jinan improved by 0.01150, with interval $[-0.02250, 0.04524]$ and probability value 0.5196. The offline advantage over the legacy comparator was supported in Los Angeles and inconclusive in Jinan.

Uncertainty in the comparison with the rigid anchor differed sharply between the two cities. A 10,000-replicate paired group bootstrap gave a Los Angeles improvement interval of $[-0.0675, 0.0729]$, with two-sided bootstrap probability value 0.9396. Jinan gave an interval of $[0.00469, 0.08293]$ and probability value 0.0274. The Los Angeles point estimate therefore remained directional with an interval spanning zero. The predeclared external gate required at least 2% city-macro improvement, both cities improving and no city regression. This engineering confirmation did not provide population-level inference across target cities.

2.4 Matched action contrast reduced waiting relative to dense residual control

The formal closed-loop matrix comprised four target scenarios, two seeds and seven policies, yielding 56 completed 3,600-s rollouts. The primary metric measured mean waiting over all departed vehicles, including unfinished trips. SUMO `write-unfinished` records accrued their waiting through the fixed horizon. Completed-only waiting, system vehicle-hours and completion were reported alongside this metric to show censoring and throughput trade-offs.

Anchored CFCMT achieved 149.17 s all-departed waiting, compared with 159.15 s for the H2O$^+$-style dense residual under the same information and refit budget. This reduction of 9.98 s (6.27%)

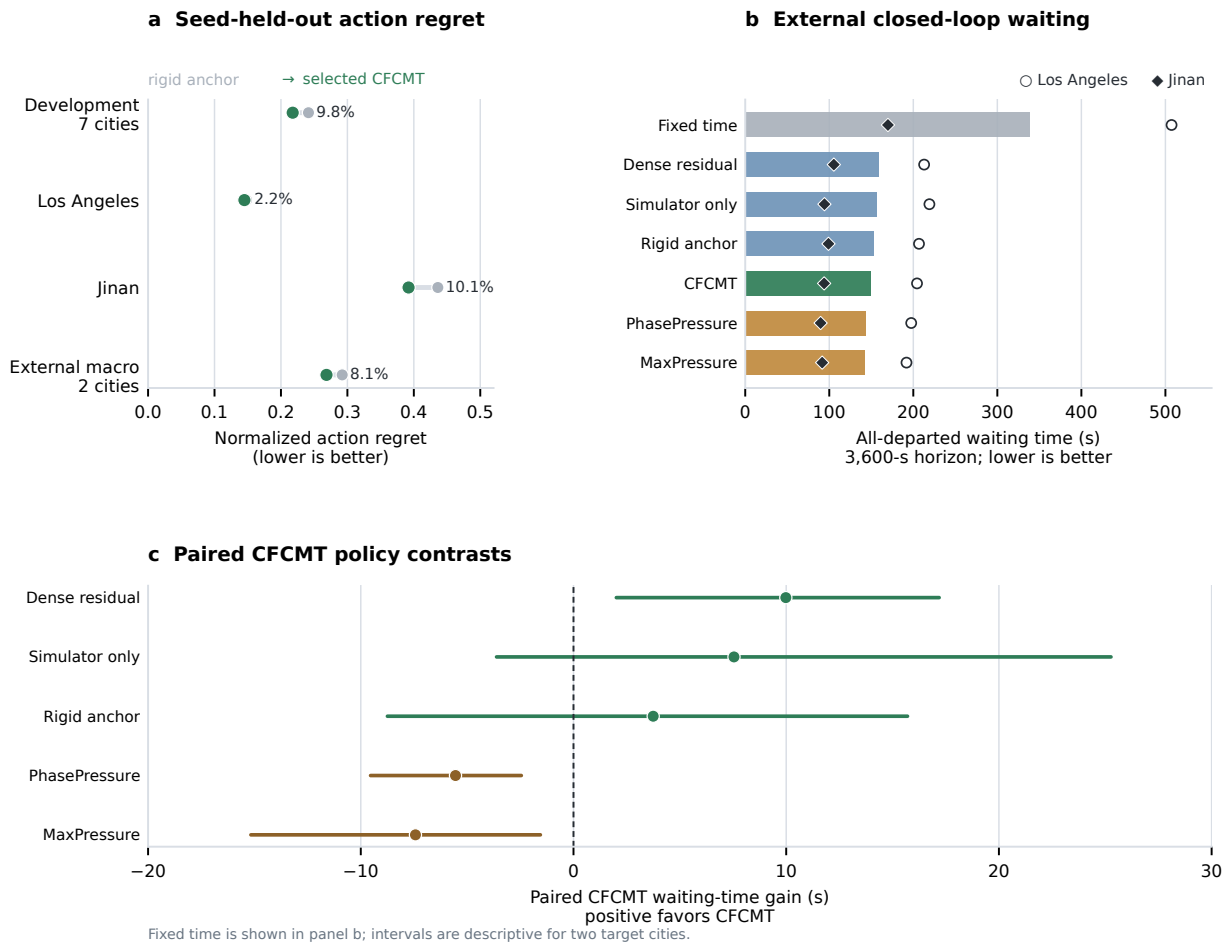


Figure 3: Original offline and seven-policy closed-loop confirmation. **a**, Seed-held-out normalized action regret for development cities and external targets. Green points show selected CFCMT scores, and grey points show the rigid anchor. **b**, All-departed waiting by policy in Los Angeles (circles) and Jinan (diamonds); bars show the equal-city macro. **c**, Paired CFCMT gains in all-departed waiting relative to the non-fixed comparators. Fixed-time results remain visible in panel **b** and Table 3. Positive values favour CFCMT; intervals are descriptive paired hierarchical bootstrap intervals for the original two-seed matrix on two target cities. The ten-new-seed confirmation is reported separately in Figure 4.

was observed in both Los Angeles (8.46 s) and Jinan (11.50 s) (Tables 3 and 10 and appendix A.3). The descriptive paired hierarchical bootstrap interval was $[2.01, 17.19]$ s, and 98.43% of replicates favoured CFCMT. Completed-only waiting also decreased from 147.47 to 133.06 s in the same comparison. System vehicle-hours decreased from 618.15 to 594.14, and completion increased from 0.870 to 0.889.

Simulator-only control and the rigid anchor achieved 156.71 and 152.91 s all-departed waiting, respectively. CFCMT improved on their point estimates by 7.54 s (4.81%) and 3.75 s (2.45%), respectively. Both intervals crossed zero, leaving the clearest closed-loop improvement against dense residual transfer (Table 4). The anchor comparison was therefore stronger in offline action regret than in closed-loop waiting.

Table 3: Original two-seed, seven-policy external 3,600-s city-macro results. Scenarios are weighted equally within each city and the two cities are weighted equally. Waiting times are seconds; system burden is vehicle-hours. "Incidents" is the equal-city mean of unique SUMO junction-collision incidents.

Policy	All-departed wait	Completed wait	System vehicle-hours	Unfinished	Completion	Incidents
Fixed time	338.58	308.72	885.18	0.133	0.661	2.00
Dense residual	159.15	147.47	618.15	0.066	0.870	0.25
Simulator only	156.71	138.68	619.66	0.069	0.877	0.00
Rigid anchor	152.91	138.40	606.30	0.072	0.882	0.50
Anchored CFCMT	149.17	133.06	594.14	0.067	0.889	0.00
PhasePressure	143.62	125.32	583.05	0.068	0.898	0.25
MaxPressure	141.74	135.42	580.29	0.065	0.896	0.25

Table 4: Original two-seed CFCMT policy contrasts for all-departed waiting. Positive gains favour CFCMT. Intervals are descriptive, not population-level city inference.

Baseline	CFCMT gain (s)	Relative gain	Descriptive 95% interval	Los Angeles gain	Jinan gain
Fixed time	189.42	55.94%	[67.11, 308.32]	302.90	75.93
Dense residual	9.98	6.27%	[2.01, 17.19]	8.46	11.50
Simulator only	7.54	4.81%	[-3.62, 25.27]	14.75	0.34
Rigid anchor	3.75	2.45%	[-8.75, 15.70]	2.36	5.14
PhasePressure	-5.55	-3.86%	[-9.54, -2.45]	-6.91	-4.19
MaxPressure	-7.43	-5.24%	[-15.17, -1.56]	-12.53	-2.33

2.5 Ten new paired seeds confirmed the frozen dense-baseline advantage

The frozen V43 CFCMT and dense models were next evaluated on ten new simulator seeds across the same four scenarios. The seed roster, primary comparison and analysis were fixed before these outcomes were opened. All 80 required 3,600-s rollouts completed with unchanged models, input networks, executor and timing; no retraining, new target labels or parameter tuning occurred.

Equal-city all-departed waiting was 150.84 s for CFCMT and 158.67 s for dense, a reduction of 7.83 s or 4.93% (Table 5 and figure 4). The preregistered paired whole-seed bootstrap gave a CFCMT-minus-dense 95% interval of $[-11.85, -4.11]$ s; the corresponding relative reduction interval was 2.64–7.31%. Nine of ten equal-city seed contrasts improved. Los Angeles improved by 5.43 s on average, with six of ten seeds favourable; Jinan's three-demand mean improved by 10.22 s, with all ten seeds favourable. Completed-only waiting decreased from 147.49 to 132.88 s, system vehicle-hours from 616.49 to 596.15, and completion increased from 87.59% to 88.66%.

The prespecified confirmation criterion was met: the primary mean difference and its 95% upper endpoint were below zero. This new-seed result is reported independently of the original 56-rollout matrix and the V91/V98 comparisons; all ten seeds were retained. The exact seed roster and unchanged execution settings are given in Section 4.12 and appendix A.4.

Table 5: Independent ten-new-seed confirmation of the frozen V43 models. Waiting includes every departed vehicle, including unfinished trips. Jinan's three demands are weighted equally, followed by equal city and seed weights. Differences are CFCMT minus dense; negative waiting differences favour CFCMT. Seed counts refer to lower all-departed waiting, and secondary outcomes use the same aggregation. Completion differences are in percentage points (pp).

Aggregation	CFCMT wait (s)	Dense wait (s)	Difference (s)	Reduction (%)	Improved seeds
Equal-city mean	150.84	158.67	-7.83	4.93	9/10
Los Angeles	207.64	213.07	-5.43	2.55	6/10
Jinan	94.05	104.27	-10.22	9.80	10/10
Secondary outcomes, equal-city means					
Metric	CFCMT	Dense	Difference		
Completed-only wait (s)	132.88	147.49	-14.61		
System vehicle-hours	596.15	616.49	-20.34		
Completion (%)	88.66	87.59	+1.08 pp		

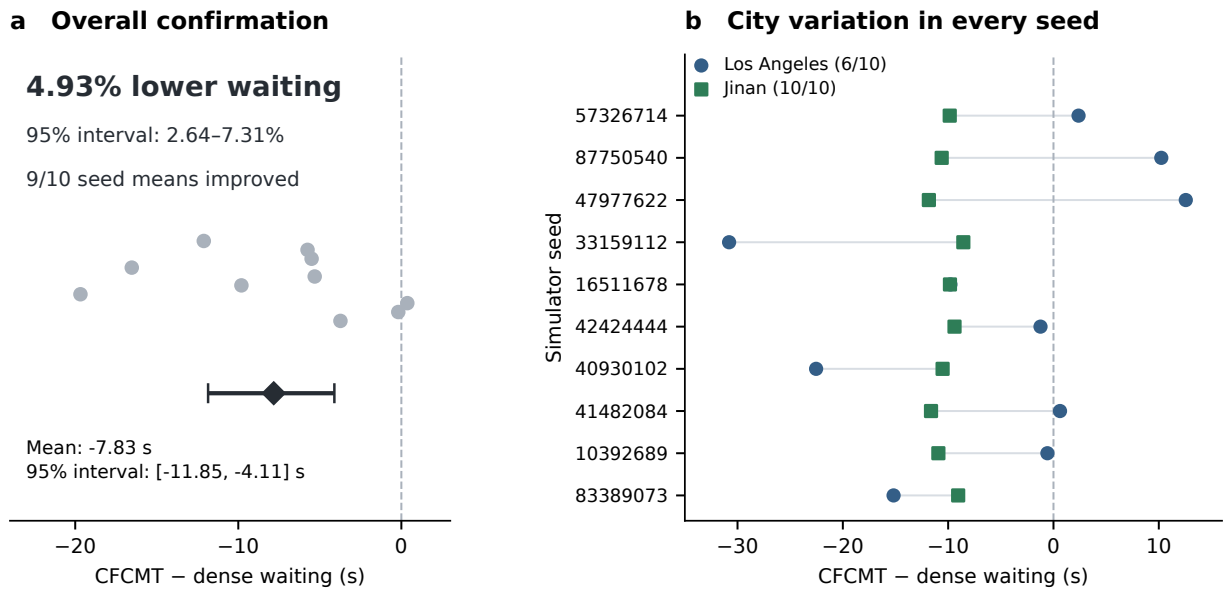


Figure 4: Independent confirmation on ten new paired simulator seeds (80 rollouts). **a**, Grey points show equal-city CFCMT-minus-dense waiting differences; vertical offsets aid visibility. The diamond and whiskers show the mean and registered 95% percentile interval from 20,000 paired whole-seed-block bootstrap replicates, retaining both policies on all four scenarios. **b**, City-specific differences for the same ten seeds. Negative values favour CFCMT. Relative reduction is computed from the ratio of macro means (Section 4.12); the interval describes seed variation within the two fixed networks.

Table 6: Closed-loop source-contribution ablation. Both policies share target adaptation groups, features, the action executor and target scenarios. The legacy comparator labelled "target only" has target weight numerically one and retains a source prior for exact ties. Positive differences denote higher all-departed waiting for CFCMT relative to that comparator. V90 is post-hoc; V91 is the preregistered fresh-seed confirmation. Incident entries report CFCMT/target only counts for their respective evaluation matrices.

Evidence	Seeds	Paired cells	Legacy near-target-only wait	CFCMT wait	CFCMT minus legacy wait	Los Angeles difference	Jinan difference	Improved	Incidents CFCMT/legacy
V90 post-hoc	2	8	149.04	149.17	+0.082%	+1.212%	-2.292%	7/8	0/1
V91 fresh (rejected)	64	256	149.09	155.14	+4.254%	+7.865%	-3.045%	29/64	25/15

2.6 A fresh near-target-only comparison exposed network-dependent transfer

The seven-policy V43 matrix omitted the frozen legacy near-target-only comparator described in the offline evaluation. We added it to the same eight scenario–seed cells after observing the main outcomes. This post-hoc V90 ablation left both models unchanged, preserving the comparator's retained source prior for exact ties. CFCMT achieved 149.17 s city-macro all-departed waiting, compared with 149.04 s for the comparator, a 0.12-s disadvantage. Los Angeles favoured the comparator by 2.45 s, whereas Jinan favoured CFCMT by 2.20 s. The post-hoc timing made this comparison descriptive rather than confirmatory.

We then froze the V91 protocol before generating 64 further simulator seeds, disjoint from all V43, V89 and V90 roles. Each seed paired CFCMT with the legacy comparator on Los Angeles and all three Jinan demands. This produced 256 paired cells and 512 completed rollouts for the fresh confirmation. The primary relative difference averaged scenarios within city, weighted the two cities equally within seed, then averaged seeds. Negative values indicated lower all-departed waiting for CFCMT than for the comparator.

The comparator achieved 149.09 s all-departed waiting, compared with 155.14 s for CFCMT. The equal-seed relative difference was +4.254% (paired 95% interval $[-0.775, 13.211]$%). Only 29/64 seeds improved, with a one-sided exact sign-test probability of 0.809. City effects had opposite signs. Jinan improved by 3.045%, with all 64 seed-level city contrasts favouring CFCMT. Los Angeles regressed by 7.865%, with improvement in 22/64 seed-level city contrasts.

The worst seed regressed by 267.56%; collision incidents totalled 25 for CFCMT and 15 for the comparator. Both policies completed all rollouts with zero teleports in this confirmation. The predeclared mean, interval, both-city, worst-seed, improved-fraction, sign-test and collision gates did not jointly pass. Fresh source-contribution confirmation was rejected (Table 6). City and seed distributions localized the rejection to the Los Angeles tail (Figure 5 and appendix A.5).

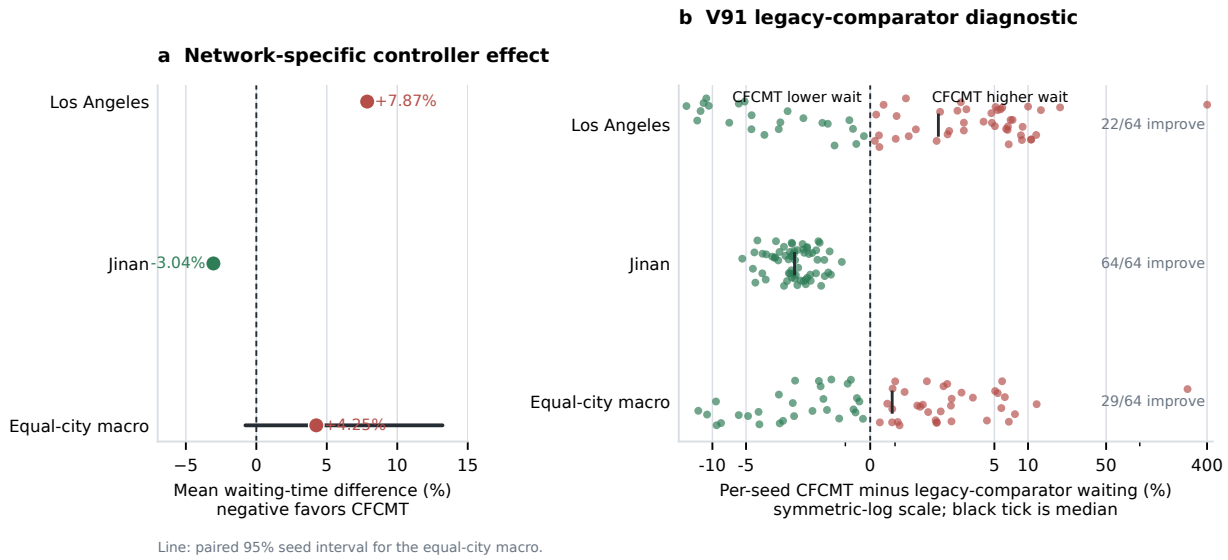


Figure 5: Network dependence in the fresh V91 comparison. **a**, Mean all-departed waiting differences for Los Angeles, Jinan and the equal-city primary estimand. Negative values favour CFCMT; the line shows the paired 95% interval for the equal-city mean. **b**, Seed-level city and equal-city differences for all 64 fresh seeds. Every Jinan city contrast favours CFCMT, whereas Los Angeles contains a long regression tail. The comparator is the frozen legacy near-target-only arm with a retained source prior. It therefore differs from a strict zero-source-row target estimator.

2.7 A target-offline selector retained a Jinan-specific controller-pair gain

After the V91 outcomes, the V98 experiment refitted target models with repaired topology context and separate source-city components. Five scenario-and-seed-blocked folds evaluated seven source groups and source masses in $\{0, 0.25, 0.5, 0.75, 1\}$. Selection used only a balanced subset of 100 target adaptation action groups (B100). Failure of the offline improvement and fold-stability gates returned the exact target-only model. Before fresh closed-loop evaluation, the selector returned source mass zero for Los Angeles. It selected Hangzhou with source mass one for Jinan before its rollouts.

The selected Jinan controller was compared with a strict zero-source-row target model on 56 further seeds. Three demand scenarios and two policies produced 336 completed rollouts in this confirmation. Equal-scenario all-departed waiting decreased by 4.527%, with a paired-seed 95% bootstrap interval of 4.279–4.787%. All 56 seeds improved, and the two-sided paired Wilcoxon probability was 7.55×10^{-11} (Table 7 and appendix A.6).

This extension used topology-repaired models and a pressure guard shared by both arms. The preselected controller improved reproducibly relative to its declared comparator under the frozen V98 protocol. The zero-source-row target estimator had lower capacity, so the contrast did not isolate source-row contribution or establish superiority over PhasePressure. V98 and V91 were reported separately because their model and guard configurations differed.

Table 7: Jinan-specific controller comparisons under two separate frozen protocols. V91 compares the original V43 controller with its legacy near-target-only arm. V98 compares the subsequently selected controller with a strict zero-source-row target model of lower capacity. Results remain separate, and the V98 contrast measures controller-pair performance without isolating source-row contribution.

Experiment	Selection/evaluation	Seeds	Improved	Mean relative waiting difference	95% interval	Comparator	Guard
V91 Jinan city diagnostic	frozen two-city protocol; Jinan diagnostic	64	64/64	-3.045%	–	final V43 legacy near-target-only	none in final V43 controller
V98 Jinan controller-pair confirmation	B100 target-offline preselection; fresh Jinan confirmation	56	56/56	-4.527%	[-4.787, -4.279]%	strict zero-source-row lower-capacity target model (causal_target_only_v2)	post-V91 shared inherited Jinan pressure guard (risk_multiplier=0.5)

Negative differences favour the first-listed controller. The V91 row is its Jinan city diagnostic and has no city-specific interval in that audit. V98 compares a preselected controller with its declared lower-capacity zero-source-row target comparator; the two rows are not pooled, and V98 does not isolate source-row contribution.

2.8 Classical pressure remained the performance ceiling

In the original external matrix, MaxPressure and PhasePressure achieved 141.74 and 143.62 s all-departed waiting, respectively. They outperformed CFCMT by 7.43 and 5.55 s, respectively, in the equal-city comparison. Both external cities favoured pressure control, and the descriptive intervals excluded zero. Pressure policies also achieved lower system vehicle-hours and higher completion ratios than CFCMT. The pressure reference defined the origin of the final controller's learned action contrast. It could select any feasible phase without falling back to either rule.

All 56 rollouts in this matrix completed with zero recorded teleports. Across policies, 171 raw collision events were consolidated into 13 unique junction incidents in six rollouts. CFCMT had zero incidents and never exceeded its paired comparator's incident count. The eight CFCMT rollouts provided an incident audit, without supporting a safety-certification claim. The metric population and incident consolidation rules are given in Appendix A.9.

2.9 A fresh-seed successor stress test did not close the pressure gap

After V43 was frozen, a separate successor track tested a longer-horizon latent originator with a globally rate-limited executor against PhasePressure. On Jinan, it used a target-state-conditioned 120-s action originator, one network-wide override at a time and a 120-s global switch cooldown. Its results were kept separate from the anchored all-pair controller's two-city matrix.

The legacy global executor yielded a mean relative all-departed waiting difference of -0.429% across 54 disclosed adaptive seeds. Its paired 95% interval was $[-0.818, -0.033]\%$ in this adaptive screen. Unlimited cooldown stays regressed by 0.104%, while a one-stay budget gave an inconclusive -0.242% point estimate. Both variants were rejected, leaving the unchanged legacy executor for fresh confirmation. We froze it before generating 64 new seeds, disjoint from all adaptive seeds and eight still-sealed prospective seeds.

On the 64 fresh seeds, the method–PhasePressure difference shrank to -0.043% (paired 95% interval $[-0.449, 0.370]\%$). The method improved 34/64 seeds; the one-sided exact sign-test probability was 0.354. The worst seed regressed by 5.21%, exceeding the preregistered 5% limit. Teleports were 0/0 and collision incidents 1/1, while the bootstrap, sign-test and worst-seed gates failed. The joint confirmation was therefore rejected (Appendix A.11 and table 15), leaving no confirmed superiority over PhasePressure on Jinan or across networks.

2.10 Native RL experiments verified baseline execution

Official RESCO and patched LibSignal experiments verified the execution of standard RL baselines. The RESCO adapter completed 42 supported full-budget runs under its native protocol. The LibSignal adapter completed 36 runs across deep Q-network (DQN), its independent variant (IDQN), PressLight, FRAP and MPLight. These runs retained native observations, rewards, action intervals, training loops and metrics. Their distinct protocols excluded them from the paired 56-rollout comparison above. Results and upstream exclusions are reported in Appendix A.12 and tables 16 and 17.

3 Discussion

The central result concerns the alignment between model structure and the control decision. CFCMT estimates normalized action contrasts, retains a rigid local anchor and selects an all-pair correction on complete held-out target seeds. This design reduced external offline regret and closed-loop waiting relative to a same-information dense residual. The dense comparator also receives matched action contrasts, so reference subtraction alone cannot explain the gain. The comparison supports the combined structural design rather than identifying each restriction's individual contribution.

The frozen-model confirmation strengthens this controller comparison beyond the original two-seed matrix. Ten new paired simulator seeds reproduced the dense-baseline advantage without refitting or tuning: waiting decreased by 4.93%, with a paired 95% interval of 2.64–7.31%. Completion increased while system burden decreased. The ten-seed result shows that the original 6.27% estimate was not confined to its two evaluation seeds, while the Los Angeles regressions on four new seeds expose remaining local variability (Section 2.5).

This result clarifies why the earlier full mechanism-world-model route was not retained as the final controller. MC-WM decomposed queue propagation, red accumulation, mobility, spillback and control cost. These components support diagnosis, but their prediction errors accumulate before the model answers the narrower phase-selection question. Good next-state or mechanism loss does not guarantee correct action ordering. Under limited target support, flexible latent or graph paths can also dominate the rigid signal. The final design preserves declared parents and local mechanism features, then adds an antisymmetric ranking correction with a measurable, bounded effect.

The controller comparisons show that transferred information has network-dependent value. V91 held target labels, feature semantics, model family and execution fixed. Across 64 fresh seeds, CFCMT reduced Jinan waiting by 3.045%, with all 64 seed-level city contrasts favourable. Los Angeles waiting increased by 7.865%, with only 22 of 64 contrasts favourable. The equal-city mean rejected a joint two-city source-benefit claim. These opposite effects show why source transfer needs an explicit abstention option rather than an assumption of benefit.

V98 tested that option through a separate post-V91 target-offline selector. Before its fresh rollouts, the selector returned exact target-only control for Los Angeles and admitted a Hangzhou source component at mass one for Jinan. On 56 further Jinan seeds (336 rollouts), the admitted controller reduced waiting by 4.527% against a strict zero-source-row target model. The paired 95% interval was 4.279–4.787%, and all 56 contrasts were favourable (two-sided Wilcoxon signed-rank probability value 7.5475×10^{-11}). The decisions preceded the rollouts, so the gain reflects the preselected Jinan controller comparison (Appendix A.6).

Pressure control remains the stronger operational reference on both external cities. MaxPressure and PhasePressure achieved lower system vehicle-hours and slightly higher completion than CFCMT. Queue pressure directly reflects service imbalance, giving these rules a strong local control signal. CFCMT's advantage over a dense residual supports structured transfer, while the pressure comparison

identifies where the learned controller remains weaker. The discarded pressure-guard experiments showed a similar boundary (Appendix A.10 and tables 13 and 14). Those pressure guards are absent from the final V43 policy.

3.1 Limitations

The formal external evidence contains two independent cities. Multiple Jinan demand scenarios add within-city coverage but do not add independent city units. Los Angeles and Jinan are CityFlow-derived networks converted and semantically audited for SUMO, rather than calibrated city-wide digital twins. Training counterfactuals and evaluation share the same SUMO physics family. The study therefore evaluates cross-network transfer in simulation, with field transfer left untested.

Target counterfactual labels also define the information scope. The uncalibrated target simulator reduces calibration effort, but its action outcomes remain labelled target data. The external comparison uses 165 and 911 adaptation groups and is an offline-adaptation result, not a zero-shot result. The seven-policy matrix uses two seeds; the independent frozen-model dense comparison adds ten new seeds on the same four scenarios. V91 uses 64 seeds over the same two target geometries, and V98 uses 56 further seeds on Jinan. These protocols remain separate, and their intervals describe the stated networks and demand settings. These evaluations use a fixed 3,600-s horizon and include unfinished vehicles. Longer incidents, non-recurrent demand and detector failure remain untested.

Source-row attribution has two specific limits. V91's comparator assigns source-labelled samples zero weight in deployed scoring, but its retained source prior can resolve exact prediction ties. Its comparator is therefore legacy near-target-only rather than strictly source-free. V98 uses a strict zero-source-row target model, but that model has lower capacity than the selected controller. Both V98 arms also share a risk multiplier of 0.5 inherited from earlier Jinan closed-loop development. The comparison establishes its declared controller-pair gain, without isolating source rows at fixed architecture or demonstrating superiority over PhasePressure.

The later source diagnostics reinforce this distinction between available headroom and a reliable selection rule. Target-seed-blocked gates either retained city regressions or selected target only throughout the seven development cities (Appendix A.7 and table 11). Mechanism-specific priors, budget-aware shrinkage, local context and state-conditioned or hierarchical utility did not resolve the source-identity controls. Forced source arms improved five of seven city means for the final target-calibrated prior, but its city interval crossed zero. The state-level utility gate that passed offline development caused a large Cologne closed-loop regression before matrix expansion. Development-only V123/V157A contrasts were superseded after a Jinan domain-handling mismatch was found; V157B corrected B100 on the reused selector but remained worse than PhasePressure. V157C retained an invalid seed in its incomplete native diagnostic, and V158/V159 failed their frozen native-prefix and endpoint-calibration gates. Their negative results and complete quantitative record are retained in Appendix A.8 and table 12. These diagnostics leave multicity source-benefit identification unresolved.

The sequential development record also matters. A diagnostic external seed exposed a one-city regression, prompting removal of an arbitrary adaptation cap and stricter complete-seed cross-fitting. That seed was permanently excluded, and a new seed authorized the final test. The final V43 cache, selectors, deployment models and closed-loop matrix were frozen before their respective next stages. This sequence is not equivalent to a single untouched preregistration (Appendices A.2 and A.13). A separate execution-layer programme's favourable 54-seed adaptive estimate shrank almost to zero on 64 newly generated Jinan seeds. It failed its preregistered bootstrap, sign-test and worst-seed criteria (Appendix A.11 and table 15). Its different 120-s latent originator and global cooldown distinguish it from the V43 comparison.

Same-protocol CrossLight, X-Light, MetaLight and GESA reproductions remain a substantive comparison gap. Their published experiments differ in target interaction budgets, state and action interfaces, simulators and training objectives. A broader benchmark should implement these systems behind the same phase executor and distinguish zero labels, passive transitions and matched counterfactual actions. Native-protocol scores cannot provide those paired comparisons (Appendix A.12).

Future work should freeze the link between source compatibility and action ranking before evaluation on additional independent cities. A calibrated digital twin or prospective signal intervention could test whether the pairwise advantage survives simulator mismatch. Passive detector or trajectory records could adapt state representations, but do not label unexecuted counterfactual actions. Objectives involving multimodal priority, spillback risk, emissions or network-level coordination may reveal value beyond a pressure score. The present evidence supports matched action contrasts relative to dense residual transfer and a separately confirmed Jinan controller-pair gain. The broader question is when source mechanisms improve control beyond both a target-only model and a strong local rule.

4 Methods

4.1 Problem definition and information budget

Let c denote a city group, n a network, g a matched action group, s_g its complete simulator state and $\mathcal{A}_g$ the feasible green stages of the focal signal. The phase executor constructs this set from the original SUMO `tlLogic` and enforces minimum green, yellow and all-red transitions. The learner requests a feasible stage, which the executor translates into the corresponding signal sequence.

The external evaluation uses target offline adaptation. Available information comprises (i) a frozen source counterfactual bank, (ii) target network, demand and signal files, (iii) target counterfactual action groups from an uncalibrated SUMO model, and (iv) target state observed during deployment. The simulator parameters are not calibrated against target field data, but item (iii) provides labelled target transitions before deployment. The experiment therefore evaluates target offline adaptation rather than zero-shot transfer. Passive traffic observations without action outcomes would provide a weaker information budget than the one evaluated here.

4.2 Matched counterfactual action groups

After a 60-s warm-up, we save the complete SUMO simulation and all signal-executor states. Each focal action $a \in \mathcal{A}_g$ starts from the same exactly restored state and is applied for one 10-s control interval. The focal signal then follows PhasePressure for five additional intervals. All other signals follow PhasePressure throughout the branch. Candidate cost is the mean number of active or pending vehicles per controlled lane over 60 one-second samples, with lower values preferred:

$$Y_g(a) = \frac{1}{H} \sum_{\tau=1}^{H} \frac{N_{\text{active}}(\tau) + N_{\text{pending}}(\tau)}{L_n}, \qquad H = 60. \tag{1}$$

Here $\tau \in \{1, \ldots, H\}$ indexes one-second samples, $N_{\text{active}}(\tau)$ and $N_{\text{pending}}(\tau)$ are the active and insertion-pending vehicle counts, respectively, and L_n is the number of controlled lanes in network n. Thus Equation (1) uses a dimensionless sample count as its summation limit. Including pending vehicles prevents a phase from being rewarded for delaying insertion. The branch collector uses `libsumo`, checks replay determinism, fails on teleports, and censors an entire matched group when

any branch has a junction collision. Group-level censoring preserves each retained group's complete candidate set.

A generalized pressure policy π_r, selected using source-only closed-loop cost, supplies one reference action $r_g = \pi_r(s_g)$. For the frozen external models, its specification is `gp_d0_o0p02_s0`. Let $Q_{0.75}(\cdot)$ denote the empirical 75th percentile. The implementation uses the numerical floor η_g, within-group scale σ_g and normalized contrast $z_g(a)$ defined jointly by

$$\begin{aligned} \eta_g &= \max\left\{10^{-8}, 10^{-6} Q_{0.75}\left(\{|Y_g(a)| : a \in \mathcal{A}_g\}\right)\right\}, \\ \sigma_g &= \max\left\{\max_{a\in\mathcal{A}_g} Y_g(a) - \min_{a\in\mathcal{A}_g} Y_g(a), \eta_g\right\}, \\ z_g(a) &= \frac{Y_g(a) - Y_g(r_g)}{\sigma_g}. \end{aligned} \tag{2}$$

The reference has $z_g(r_g) = 0$. For a selected action $\widehat{a}_g$, evaluation uses the normalized action regret in

$$\mathrm{Reg}_g(\widehat{a}_g) = \frac{Y_g(\widehat{a}_g) - \min_{a\in\mathcal{A}_g} Y_g(a)}{\sigma_g}. \tag{3}$$

All reported offline regret values use Equation (3).

Assumption 1 (action-common domain error)**.** *At one restored state, the candidate cost admits the decomposition*

$$Y_g(a) = b_g + q_g(a) + e_g(a), \tag{4}$$

where b_g is a state-specific term shared by every candidate, $q_g(a)$ is the action-dependent component intended to transfer, and $e_g(a)$ is the remaining action-dependent error.

Proposition 1 (action-common cancellation and rank invariance)**.** *Under Assumption 1, b_g cancels exactly from every candidate-versus-reference numerator. Adding any state-specific constant to all candidate outcomes leaves the action ordering and regret minimizer unchanged. It also leaves the numerical values of $z_g(a)$ unchanged whenever the within-group range, rather than the relative floor η_g, determines σ_g.*

Proof. For any $a \in \mathcal{A}_g$, subtracting the reference outcome in Equation (4) gives $Y_g(a) - Y_g(r_g) = q_g(a) - q_g(r_g) + e_g(a) - e_g(r_g)$, so b_g cancels. A common additive constant cancels from every difference and does not change the raw within-group range. Because $\sigma_g > 0$, any change in its numerical floor rescales all contrasts by the same positive factor and cannot change their ordering. If the range is the active term in Equation (2), σ_g itself is unchanged, which also preserves every normalized value. □

This proposition motivates the estimand but does not remove action-dependent simulator error. Exact state restoration is therefore a protocol requirement, not merely a variance-reduction device.

4.3 Parent-restricted rigid anchor

Let θ denote the fitted anchor parameters and define $A_g(a) := A_\theta(s_g, a)$. The anchor directly regresses $z_g(a)$ for non-reference candidates and assigns zero to the reference. Its 29 declared parents comprise local total queue, vehicle count, speed and occupancy; candidate and reference green/red queue; downstream queue and occupancy; service and red pressure; switch and clearance indicators; and current-phase overlap. City identifiers, static target context and unrestricted graph embeddings

are excluded from the direct predictor. The parent set encodes a predefined modelling hypothesis about local action mechanisms rather than a structure inferred from field observations.

The estimator is a `HistGradientBoostingRegressor` with 100 boosting iterations, at most 15 leaves, minimum leaf size 12 and ℓ_2 regularization 10. Each city domain receives equal total weight, each matched action group receives equal weight within a domain, and negative versus non-negative candidate contrasts are reweighted so the dominant sign does not determine the fit. Targets are normalized independently by σ_g. The action-level anchor uncertainty $u_{A,g}(a)$ is calibrated from the 90th-percentile absolute residual on source leave-one-domain predictions.

4.4 Antisymmetric all-action correction

The correction model uses every unordered pair $\{a_i, a_j\} \subset \mathcal{A}_g$, with $i \neq j$ indexing candidate actions. Let x_g contain 23 action-common state parents and $\phi_g(a)$ contain 40 phase-local action parents. With square brackets denoting vector concatenation, the training design d_{gij} and normalized pair target t_{gij} are

$$d_{gij} = [x_g,\, \phi_g(a_i) - \phi_g(a_j)]\,, \qquad t_{gij} = \frac{Y_g(a_i) - Y_g(a_j)}{\sigma_g}. \tag{5}$$

Every row defined in Equation (5) is augmented by its reverse $(d_{gji}, -t_{gij})$. Let ψ denote the fitted pairwise-model parameters. At prediction, the regressor f_ψ is explicitly antisymmetrized,

$$\widehat{\delta}_{gij} = \tfrac{1}{2}\left[f_\psi(x_g, \phi_g(a_i) - \phi_g(a_j)) - f_\psi(x_g, \phi_g(a_j) - \phi_g(a_i))\right], \tag{6}$$

and converted first to the unshifted all-pair score $\widetilde{P}_g$ and then to the reference-shifted score $P_g(a) := P_\psi(s_g, a)$:

$$\begin{aligned} \widetilde{P}_g(a_i) &= \frac{1}{K_g - 1} \sum_{j \neq i} \widehat{\delta}_{gij}, \quad K_g = |\mathcal{A}_g|, \\ P_g(a_i) &= \widetilde{P}_g(a_i) - \widetilde{P}_g(r_g). \end{aligned} \tag{7}$$

Thus $P_g(r_g) = 0$. Pair-cycle inconsistency augments the 90th-percentile residual in the action-level pairwise uncertainty $u_{P,g}(a)$.

Proposition 2 (antisymmetry and ranking recovery)**.** *Equation* (6) *gives* $\widehat{\delta}_{gji} = -\widehat{\delta}_{gij}$. *If* $\widehat{\delta}_{gij} = z_g(a_i) - z_g(a_j)$ *for every pair, then*

$$\begin{aligned} \widetilde{P}_g(a_i) &= \frac{K_g}{K_g - 1}\left[z_g(a_i) - \overline{z}_g\right], \\ P_g(a_i) &= \frac{K_g}{K_g - 1}\left[z_g(a_i) - z_g(r_g)\right]. \end{aligned} \tag{8}$$

Here $\overline{z}_g = K_g^{-1} \sum_{a \in \mathcal{A}_g} z_g(a)$. *The positive scale preserves* $\arg\min_a z_g(a)$.

Proof. Swapping i and j in Equation (6) negates the expression. Under the stated exact-pair condition, summing in Equation (7) yields Equation (8); subtracting a common mean and multiplying by a positive constant cannot change the minimizer. □

The pairwise model uses the same gradient-boosting family as the anchor. Its state parents comprise local aggregate state, time, queue concentration and signal-graph neighbourhood summaries; its action parents include phase duration, queue, occupancy, downstream load, service pressure, transition and graph-action summaries. Context values are used only to calibrate support and uncertainty and are excluded from the direct prediction features.

4.5 Anchored interpolation and local cap

For one action group, the deployed score is

$$S_g(a) = (1-\alpha_g)A_g(a) + \alpha_g P_g(a), \qquad 0 \le \alpha_g \le 1. \tag{9}$$

Candidate selectors include constant choices $\alpha_g \in \{0, 0.1, \ldots, 1\}$ and local confidence-capped weights. For the local form, for any action-indexed vector v_g define $\mathrm{range}(v_g) = \max_{a\in\mathcal{A}_g} v_g(a) - \min_{a\in\mathcal{A}_g} v_g(a)$. Let

$$\begin{aligned} a_{A,g} &= \arg\min_{a\in\mathcal{A}_g} A_g(a), & a_{P,g} &= \arg\min_{a\in\mathcal{A}_g} P_g(a), \\ h_g &= \max\{\mathrm{range}(A_g), \mathrm{median}_a\, u_{A,g}(a), 10^{-6}\}, \\ D_g &= \mathrm{range}(P_g - A_g)/h_g, \\ M_g &= \max\{P_g(a_{A,g}) - P_g(a_{P,g}), 0\}/h_g, \\ U_g &= \{u_{P,g}(a_{A,g}) + u_{P,g}(a_{P,g})\}/h_g. \end{aligned} \tag{10}$$

Here h_g is the local score scale, D_g the normalized score disagreement, M_g the pairwise preference margin and U_g its uncertainty sum. With perturbation cap κ and confidence multiplier ζ, the diagnostics in Equation (10) determine the local weight

$$\alpha_g = \min\{1, \kappa/\max(D_g, 10^{-12})\}\mathbf{1}\{M_g \ge \zeta U_g\}, \tag{11}$$

where $\mathbf{1}\{\cdot\}$ is the indicator function.

Proposition 3 (blend and decision-error bounds)**.** *For a group-constant α_g, if the non-negative constants ε_A and ε_P bound the anchor and pairwise uniform score errors, respectively, so that $\|A_g - z_g\|_\infty \le \varepsilon_A$ and $\|P_g - z_g\|_\infty \le \varepsilon_P$, then*

$$\|S_g - z_g\|_\infty \le (1-\alpha_g)\varepsilon_A + \alpha_g\varepsilon_P. \tag{12}$$

For $\widehat{a}_g = \arg\min_{a\in\mathcal{A}_g} S_g(a)$ and $a_g^* = \arg\min_{a\in\mathcal{A}_g} z_g(a)$,

$$z_g(\widehat{a}_g) - z_g(a_g^*) \le 2\|S_g - z_g\|_\infty. \tag{13}$$

Here $\|v_g\|_\infty = \max_{a\in\mathcal{A}_g} |v_g(a)|$.

Proof. The error bound in Equation (12) follows from convexity of the sup norm. The decision bound in Equation (13) follows by adding and subtracting $S_g(\widehat{a}_g)$ and $S_g(a_g^*)$, using the minimizing property of $\widehat{a}_g$, and bounding each remaining discrepancy by $\|S_g - z_g\|_\infty$. □

These are deterministic conditional bounds, not guarantees that either fitted error is small in a new city.

Corollary 1 (bounded local perturbation)**.** *Whenever the confidence indicator in Equation* (11) *passes,*

$$\mathrm{range}(S_g - A_g) = \alpha_g\,\mathrm{range}(P_g - A_g) \le \kappa h_g. \tag{14}$$

When it fails, $S_g = A_g$. Thus Equation (14) *bounds score perturbation relative to the anchor, but it is not a queue-stability or safety guarantee.*

4.6 Seed-blocked target selection and full refit

For each external city, we retain all valid groups from adaptation seeds 5057 and 6067. In fold one, both models are fitted on the source bank and target seed 5057; target seed 6067 evaluates every candidate α_g. Fold two reverses the target seeds. An admissible candidate must match or improve the anchor's mean OOF regret and regress by at most 0.1 in either complete-seed fold. The selector minimizes mean OOF regret among admissible candidates, with deterministic tie breaking. Los Angeles selects $(\kappa, \zeta) = (1, 0.25)$ and Jinan selects constant $\alpha_g = 0.9$.

After the choice is frozen, the model is fitted once on the source bank plus all target adaptation groups: 165 groups for Los Angeles and 911 for Jinan. The OOF models are discarded. CFCMT and the learned baselines share the same target group identifiers and full-refit call. Serialized models are reloaded and checked before deployment. The seed-8171 evaluation cache is required to be absent at the joint freeze and the diagnostic-only seed 7079 is permanently excluded from V43 selection and confirmation.

4.7 Closed-loop controller and comparators

At each 10-s decision time, the runtime enumerates every currently feasible phase, evaluates its frozen 60-s action score and requests the phase with the minimum deployed score $S_g(a)$. The phase executor enforces physical transition timing. No uncertainty guard, pressure fallback or target closed-loop tuning is active in the final V43 policy. The pressure reference is recomputed only to create the zero coordinate required by the contrast model.

The same-protocol comparators are fixed time, MaxPressure, PhasePressure, simulator-only contrast MPC, the rigid anchor alone and a dense residual. The dense residual receives the same matched action contrasts and refit budget but allows unrestricted feature/context mixing in a dense residual world model. It is labelled H2O$^+$-style because it combines a simulator with a learned residual. It does not reproduce the original H2O$^+$ algorithm, optimizer or online training loop. Because both methods use matched action contrasts, subtracting the reference action alone cannot explain any CFCMT advantage.

Official RESCO and LibSignal controllers retain their upstream state, reward, timing and training definitions. These runs assess execution under the native protocols rather than the paired CFCMT comparison.

4.8 Legacy near-target-only source-contribution test

The original source-contribution test uses the pre-existing `causal_target_only` model family. This comparator has the same causal action-advantage features, target domain and target adaptation groups as the corresponding baseline artifact. It sets prior group strength to zero and the maximum target weight to one. Both external artifacts had more than four target groups and selected target weights numerically equal to one. A source submodel is still fitted and retained; its coefficient in the deployed mean, uncertainty and trust predictions is zero, but its prior can resolve exact action ties. We therefore designate this frozen arm the legacy near-target-only comparator. Its retained tie-breaking prior distinguishes it from a strict zero-source-row estimator.

The first deployment of this frozen comparator, V90, reused the eight V43 scenario–seed cells and was specified only after the original seven-policy outcomes were known. V90 is therefore post-hoc. The subsequent V91 protocol was frozen before any V91 outcome, with neither model refitted. A deterministic generator produced 64 seeds disjoint from every seed-valued field in V43 and V89 and from the two V90 closed-loop seeds. Four scenarios and two policies gave 512 required rollout identities, all of which completed.

For policy π, city c, demand scenario m in the city-specific set $\mathcal{M}_c$ and seed index k, let $W^{\pi}_{c,m,k}$ be the all-departed mean waiting time. With the external city set $\mathcal{C} = \{\text{Los Angeles}, \text{Jinan}\}$, the scenario and equal-city means are

$$\bar{W}^{\pi}_{c,k} = |\mathcal{M}_c|^{-1} \sum_{m \in \mathcal{M}_c} W^{\pi}_{c,m,k}, \qquad \bar{W}^{\pi}_{k} = |\mathcal{C}|^{-1} \sum_{c \in \mathcal{C}} \bar{W}^{\pi}_{c,k}. \tag{15}$$

Using the aggregation in Equation (15), the paired relative waiting effect is

$$\rho_k = \frac{\bar{W}^{\mathrm{CFCMT}}_{k} - \bar{W}^{\mathrm{legacy}}_{k}}{\bar{W}^{\mathrm{legacy}}_{k}}, \tag{16}$$

where $\pi = \text{legacy}$ denotes the frozen legacy near-target-only comparator and negative values favour CFCMT. The primary estimate is the mean of ρ_k over 64 seeds, with 20,000 paired-seed bootstrap replicates. The predeclared joint gate required a non-positive mean and interval upper bound, non-positive means in both cities, at most 5% worst-seed regression, at least 50% improved seeds, one-sided exact sign-test probability value $p_{\mathrm{sign}} \leq 0.05$, no excess teleports or collision incidents, all identities and no failed-seed exclusion. Failure of any component rejects the source-contribution confirmation.

4.9 Post-V91 target-offline conditional source selection

V98 is a separate post-V91 successor and is not part of the frozen V43 method. Target models were refitted after repairing topology context, with one candidate component for each of seven source-city groups. Five scenario-and-seed-blocked folds evaluated source masses $\{0, 0.25, 0.5, 0.75, 1\}$ on a balanced B100 subset of target counterfactual groups. A source component was admitted only when its mean normalized-regret improvement exceeded 0.5%, its one-sided 95% upper confidence bound was below zero and its worst fold regressed by less than 1%. Failure of any gate selected the exact target-only model. This rule returned source mass zero for Los Angeles and Hangzhou source mass one for Jinan before the fresh confirmation rollouts were observed.

The strict comparator uses the corrected `causal_target_only_v2` estimator and fits target rows only. The fresh Jinan confirmation used 56 deterministic seeds produced by NumPy's 64-bit Permuted Congruential Generator (PCG64), excluded from V93 and the earlier V98 confirmation. Three demand scenarios and two policies gave 336 completed rollouts. Waiting was averaged equally across the three scenarios within each seed; uncertainty used a paired-seed bootstrap, and the paired Wilcoxon statistic was computed over the 56 seed means.

The selector uses no target closed-loop outcome, although the full V98 protocol incorporates earlier closed-loop development. Both Jinan arms share a pressure-guard risk multiplier of 0.5 inherited from that development. The strict zero-source-row comparator also has lower model capacity than the source-aware estimator. V98 therefore evaluates the frozen conditional protocol without isolating source-row contribution at fixed architecture.

4.10 Separate post-freeze execution-layer stress test

The V86–V89 track is a distinct successor experiment on `jinan_3x4_real`. Its originator predicts a 120-s target-simulator cost from target-state and action features; the executor permits at most one network-wide override and requires at least 120 s between effective phase switches. The method was selected only from 54 disclosed adaptive seeds. Unlimited cooldown stays and a one-stay-per-cooldown variant were evaluated on the same adaptive partition and could not replace the legacy executor.

Before any V89 rollout, a deterministic generator selected 64 new seeds after excluding the 54 adaptive seeds and eight separately sealed V85 seeds. Each seed paired the frozen executor with PhasePressure under the same target scenario. The preregistered joint rule required non-positive mean relative waiting difference, a non-positive paired-bootstrap 95% upper bound, no more than 5% worst-seed regression, at least 50% improved seeds, one-sided exact sign-test $p_{\text{sign}} \leq 0.05$, at least 75% active seeds, no aggregate excess in teleports or collision incidents, closed phase accounting and all 128 rollout identities. Twenty thousand paired bootstrap replicates were fixed in the protocol. No failed seed was excluded, and V85 remained sealed. This track is reported as a stress test and is not part of V43 method selection.

4.11 Networks, splits and experimental execution

Development uses 18 networks in seven groups. Source model seeds are 2027 and 3037 and the excluded development evaluation seed is 4047. The successful development stage uses up to 100 target-style groups inside each held-out source-city exercise. External confirmation uses one Los Angeles network and one Jinan geometry with three demand files. Jinan scenarios are weighted equally before city aggregation, preventing that city from receiving triple the weight of Los Angeles.

The original closed-loop matrix runs four scenarios, two seeds and seven policies for 56 paired rollouts. Every rollout uses SUMO 1.22.0, `libsumo` and a 3,600-s simulation horizon with a 60-s warm-up. The experiment was parallelized across cluster workers, while numerical libraries were restricted to one thread per process to avoid nested oversubscription. Each rollout records its result and `tripinfo` hashes, source snapshot identity, model identity, SUMO version and metric population.

The fresh source-contribution confirmation uses the same four scenarios, simulator version, warm-up, horizon, action executor and metric implementation. Only the two frozen policies and 64 fresh seeds differ, producing 512 rollouts. Six process-isolated shards ran on six nodes with 22 workers per node. Full rollout records remain on the execution host; six complete shard summaries and the launch and audit manifests are materialized with the manuscript evidence.

4.12 Independent confirmation of the frozen V43 dense comparison

After the original two-seed results, a separate protocol fixed ten new seeds and the analysis before any new rollout. Python's Mersenne Twister generator, initialized with 20261022, drew unique integers from 10,000,000 through 99,999,999, excluding 567 seed values recorded in 159 existing configuration files, including used and reserved seeds. The resulting roster is given in Appendix A.4. Each seed paired the original CFCMT and dense artifacts on Los Angeles and all three Jinan demands, producing 80 required rollouts. The original source snapshot, converted networks, SUMO version, models and selected blends were retained, as were the sparse executor, 10-s control interval, 60-s prediction horizon, warm-up and simulation end time. There was no refitting, additional label collection, model selection or parameter tuning.

The primary contrast was the equal-city mean waiting difference, CFCMT minus dense, in seconds per departed vehicle. Within each seed, the three Jinan demands were averaged equally, followed by equal city weights and then equal weights across the ten seeds. The relative contrast divided the difference of the two macro means by the dense macro mean; it did not average seed-level percentages. Twenty thousand paired percentile bootstrap replicates used PCG64 initialized with 20261022. Each draw resampled ten complete seed blocks, preserving both policies on all four scenarios together and keeping cities fixed. Confirmation required all 80 identities to be valid and complete, the primary mean difference to be negative and its two-sided 95% upper endpoint to be

below zero. Missing or failed runs could not be replaced by new seeds or encoded as zero. Secondary metrics and both city effects were retained, with no pooling with the original matrix, V91 or V98.

4.13 Outcomes, aggregation and uncertainty

The preregistered closed-loop primary is mean vehicle waiting time. The implementation computes it over all departed vehicles, including unfinished `tripinfo` records whose waiting and duration are accrued to the fixed horizon. We additionally report completed-only waiting, system vehicle-hours, unfinished fraction, completion ratio, throughput, mean queue per lane, teleports, emergency stops and junction collisions. Together, these outcomes account for withheld departures and unfinished trips that could otherwise make a controller appear effective. Trip-based waiting includes vehicles departing during warm-up. Queue and system-burden samples begin at the end of warm-up. Completion is the number of arrived vehicles divided by departed vehicles plus vehicles still pending departure at the horizon.

Scenario-seed results are first averaged within scenario, scenarios are averaged equally within city and the two external cities are averaged equally. All policy contrasts remain paired on city, scenario and seed. Offline uncertainty uses 10,000 paired action-group bootstrap replicates. For the original seven-policy matrix, uncertainty uses 10,000 paired hierarchical replicates that sample city, then scenario, then seed. With only two independent target cities, these intervals describe sensitivity of the observed matrix; they are not estimates of a general population of cities. No multiplicity-adjusted claim or universal city-level significance is made. The ten-new-seed dense confirmation instead uses the whole-seed-block procedure in Section 4.12; its interval describes seed variation within the two fixed target networks.

The V91 source-contribution interval instead resamples the 64 paired seed statistics in Equation (16) 20,000 times. The exact sign test uses the number of seeds with $\rho_k < 0$. City-specific relative differences are secondary diagnostics because the independent city count remains two.

Raw SUMO collision events are consolidated by participant, lane and time into unique junction incidents. Teleports are a hard validity failure; collisions are retained as outcomes in closed loop and compared pairwise. The original matrix requires all 56 authorized records, exact provenance identities, complete population accounting, readable `tripinfo` hashes and zero teleports. The independent dense confirmation requires all 80 registered identities, unchanged models and execution settings, complete vehicle accounting and zero teleports; every seed remains in the report. The V91 integrity audit analogously requires all 512 frozen identities and retains every seed regardless of performance or incident status.

4.14 Meaning of "causal"

CFCMT follows two causal design commitments. Candidate actions intervene from a common restored simulator state, and direct predictors are restricted to declared local parent sets. Proposition 1 is algebraic under Assumption 1, while Propositions 2 and 3 and Corollary 1 concern score construction. The study does not identify a causal graph from observational traffic, prove that the declared parents are invariant in the field, or estimate effects of signal interventions in Los Angeles or Jinan. "Causal" therefore describes the model and counterfactual simulation design, not field causal identification.

5 Conclusion

Anchored CFCMT formulates cross-network signal adaptation as normalized, matched action ranking. Its parent-restricted anchor and antisymmetric correction reduced held-out action regret in development and on two external networks. In the frozen external comparison, CFCMT reduced all-departed waiting by 6.27% relative to a same-information dense residual baseline. Ten new paired seeds across the same four scenarios independently confirmed this advantage with both models frozen: waiting decreased by 4.93%, with a paired 95% interval of 2.64–7.31% and improvement in nine of ten seed-level contrasts.

Controller gains depended on the target network. The 64-seed V91 comparison improved Jinan waiting by 3.045% but worsened Los Angeles waiting by 7.865% relative to its legacy near-target-only comparator. A separate target-offline selector subsequently returned target-only control for Los Angeles and admitted a Hangzhou source component for Jinan. Across 56 further Jinan seeds, the admitted controller reduced waiting by 4.527%, with a paired 95% interval of 4.279–4.787% and improvement on every seed. This V98 result establishes a reproducible controller-pair gain; its lower-capacity zero-source-row comparator leaves source-row attribution unresolved.

The evidence supports structured action-contrast modelling and selective target adaptation within the tested SUMO benchmarks. MaxPressure and PhasePressure remained stronger controllers, and later source-selection and native-calibration tests did not establish a general source benefit. Broader transfer claims require independent target networks and evaluation beyond the shared simulator physics.

6 Data and Code Availability

The source benchmark combines public scenarios executed with SUMO (Lopez et al., 2018), including RESCO benchmark networks (Ault and Sharon, 2021), with public traffic-signal benchmark files distributed by LibSignal (Mei et al., 2022). The external Los Angeles input is the public LibSignal `LA_1x4` benchmark (Mei et al., 2022), represented by a CityFlow roadnet and flow file (Zhang et al., 2019). The files are fixed at LibSignal repository commit `127af9f`. The Jinan inputs are the roadnet and three real-flow variants distributed with the public CoLight `Jinan/3_4` benchmark (Wei et al., 2019b), fixed at repository commit `04440b7`. Conversion to SUMO, route conservation, signal-link identity, collision-free admission and exact source hashes are recorded in the V8 conversion manifest.

The manuscript source-data directory contains comma-separated values (CSV) files for every new figure and table. A manifest links the development audit, source and adaptation caches, deployment models, held-out result and closed-loop aggregate through Secure Hash Algorithm 256-bit (SHA-256) digests. The source-contribution table retains the final-V43 legacy near-target-only label in the V90 and V91 audits. A separate controller-comparison table reports the V91 and V98 audits independently. Its V98 row compares a preselected controller with a lower-capacity target model fitted without source rows. The manifest also links the metadata-only audit of all 336 remote V98 result identities, which required no transfer of raw result files.

The independent ten-new-seed V43 dense confirmation has a separate frozen protocol and 80 complete rollout summaries. Its figure-generation script exports per-rollout, paired-seed, scenario, city and overall CSV tables, including the registered 20,000-replicate seed-block bootstrap intervals. The original models and raw `tripinfo` files remain in the execution archive; this confirmation adds no training data or fitted models.

The seven-city source-gate CSV and LaTeX table are regenerated from the frozen V145, V146 and V148 aggregate JavaScript Object Notation (JSON) files; all three decisions remain rejected.

The post-V148 source-remedy CSV and table use the canonical V150A, V150C, V150I–V150L, V151A–V153A and V156A–V159 result JSON files, together with the V157B bound arm manifest. They retain the failed source-identity and closed-loop decisions and mark the V123/V157A all-budget comparison as confounded by domain handling. The corrected V157B B100 pass remains limited to its reused-selector estimand. The incomplete V157C artifact retains seed 27178 as invalid at its fixed 480-s checkpoint; the remaining two seeds are reported descriptively.

The V158 artifact records the completed ten-seed native-prefix OOF failure and the unrun reserve. The V159 artifact binds the target-only direct-endpoint ablation to the unchanged V158 native B100 bank, with no new simulator trajectories. Both failed OOF decisions and the withheld reserve evaluations are retained. Their estimates and interpretation are reported in Appendix A.8.

The 512 full V91 rollout records remain in the execution archive, with six complete shard summaries available locally. A persistent public archive and its digital object identifier (DOI) have not yet been provided for the code snapshot, converted networks, configuration files, model artifacts, aggregate JSON, per-rollout results, `tripinfo` files and figure-generation scripts. Public distribution remains subject to the licences of the upstream benchmark files.

A Supplementary Methods and Results

A.1 Network inventory

Table 8: Source and external target network inventory parsed from the SUMO `net.xml` files. Jinan uses one geometry and three demand files.

Role	City group	Networks	Traffic signals	Lanes	Scenarios
Source	Atlanta	1	5	55	atlanta_1x5
Source	Cologne	3	12	252	cologne1, cologne3, cologne8
Source	Hangzhou	6	36	532	hangzhou_bc_tyc, hangzhou_kn_hz, hangzhou_qc_yn, hangzhou_sb_sx, hangzhou_4x4, hangzhou_4x4_hetero
Source	Ingolstadt	3	32	2227	ingolstadt1, ingolstadt7, ingolstadt21
Source	New York	1	196	2562	manhattan_28x7
Source	RESCO synthetic	2	32	360	grid4x4, arterial4x4
Source	Salt Lake City	2	4	88	saltlake_400s_200w_q1_weekday_peak, saltlake_state_university_q1_weekday_peak
External target	Los Angeles	1	4	157	la_1x4
External target	Jinan	1	12	186	jinan_3x4 (3 demand files)

The external CityFlow files were converted to SUMO and revised through a semantic admission audit. Phase identities use final SUMO connection indices, permissive links remain distinct from protected movements, all routes are contiguous and demand is conserved. A 3,600-s admission run had no teleports or collisions. The admitted conversion tree has SHA-256 `4a2e17618c2ac584fc767562f3a7096c95d8f904f03ef6baa9cd5d34d282eadd`.

A.2 Sequential development and confirmation

Table 9: Sequential development and confirmation of the pairwise method. Positive percentages denote lower normalized action regret than the rigid anchor. Seed 7079 is diagnostic evidence that informed V43 and is not reused as V43 confirmation.

Stage	Independent check	Result	Protocol consequence
V38 global pairwise	Salt Lake, 2 networks	−12.38% vs rigid; FAIL	reject global correction
V39 anchored selector	7-city development	−0.95% macro; FAIL	reject first anchor blend
V40 cross-fitted selector	7-city nested folds	selected anchor; FAIL	add seed-held-out evaluation
V41 seed-held-out	seed 4047, 7 cities	+9.84%, 6/7 improve; PASS	freeze for external diagnosis
V42 external diagnostic	seed 7079, 2 cities	+4.50% macro, 1/2 improve; FAIL	exclude seed; remove B100 cap
V43 external confirmation	seed 8171, 2 cities	+8.09%, 2/2 improve; PASS	authorize closed loop

V42 improved the city-macro mean but failed the predeclared requirement that both external cities improve. Los Angeles regressed by 0.00072 normalized regret. V43 then removed the arbitrary 100-group cap, used every valid adaptation group and cross-fitted by complete adaptation seed. The later seed 8171, rather than 7079, is the formal external confirmation.

A.3 City-resolved closed-loop outcomes

Table 10: External closed-loop outcomes before equal-city aggregation. Waiting is the all-departed 3,600-s metric in seconds. Incident entries are means over the two closed-loop seeds within that city and policy.

Policy	Los Angeles wait	Jinan wait	Los Angeles completion	Jinan completion	Los Angeles incidents	Jinan incidents
Fixed time	507.33	169.84	0.449	0.873	4.00	0.00
Dense residual	212.88	105.41	0.838	0.902	0.50	0.00
Simulator only	219.17	94.26	0.851	0.903	0.00	0.00
Rigid anchor	206.78	99.05	0.862	0.902	1.00	0.00
Anchored CFCMT	204.42	93.91	0.874	0.903	0.00	0.00
PhasePressure	197.51	89.73	0.890	0.905	0.50	0.00
MaxPressure	191.89	91.58	0.888	0.904	0.50	0.00

The CFCMT-versus-dense residual waiting improvements are 8.46 s in Los Angeles and 11.50 s in Jinan. CFCMT-versus-MaxPressure differences are -12.53 and -2.33 s, where negative values indicate that MaxPressure is better. The gains over the dense residual therefore do not imply gains over MaxPressure.

A.4 Ten-new-seed frozen dense-baseline confirmation

The ten simulator seeds, in their registered order, were 57326714, 87750540, 47977622, 33159112, 16511678, 42424444, 40930102, 41482084, 10392689 and 83389073. Each seed supplied eight paired records: CFCMT and dense on one Los Angeles and three Jinan scenarios. All 80 records were new runs; none reused an original result or omitted a failed identity. Model identities, execution settings and complete vehicle ledgers were independently checked. Every rollout recorded zero teleports.

The overall waiting difference was -7.826876 s, with the preregistered whole-seed-block 95% interval $[-11.851786, -4.108126]$ s (Section 4.12). The relative reduction was 4.932804%, with interval 2.635285–7.312375%. Each of the three Jinan demand means favoured CFCMT: waiting decreased by 8.91, 10.58 and 11.17 s for the nominal, 2,000 and 2,500 demand files, respectively. Los Angeles had four unfavourable seed contrasts, all retained in its mean and in Figure 4. The source-data tables retain all per-rollout and paired seed outcomes alongside the city and scenario aggregates. No new pressure-policy comparison or source-row ablation was performed in this confirmation.

A.5 Fresh source-contribution diagnostics

The comparator artifacts were created in the V43 baseline freeze, before the original closed-loop outcomes. Their deployed score assigns zero coefficient to source-labelled samples, and their selected target weights were within numerical precision of one in both cities. However, a retained source-trained prior can resolve exact prediction ties. We therefore call this the legacy near-target-only comparator rather than a strict zero-source-row arm. V90 added these artifacts to the eight original scenario–seed cells only after the main comparison and is therefore not a confirmation. V91 froze 64 unique seeds before execution and required 512 exact policy identities. All rollouts completed without seed exclusion.

The largest V91 failure occurred at seed 7417817 in Los Angeles. CFCMT waiting was 975.27 s with completion 0.270, compared with 194.74 s and completion 0.919 for the legacy near-target-only comparator; neither policy teleported or recorded a collision in that cell. This record is retained in the primary estimate. Across the 64 Los Angeles cells, the median relative difference was +2.75%, only 22 favoured CFCMT and 21 regressed by more than 5%. Across 192 Jinan scenario–seed cells, 179 favoured CFCMT, and all 64 Jinan seed-level city contrasts favoured CFCMT. The city-mean waiting effects were −3.045% in Jinan and +7.865% in Los Angeles, where negative values favour CFCMT. These diagnostics describe the opposite city outcomes; the preregistered seed-level analysis remains unchanged.

A.6 Post-V91 V98 controller-pair confirmation

V98 follows a separate post-V91 protocol and is reported independently of V91. Its target-offline selector returned exact target-only control for Los Angeles and selected a Hangzhou source component at mass one for Jinan before any fresh V98 rollout was opened. Los Angeles was therefore an ex ante abstention, not a city removed after observing the confirmation outcomes. The Jinan matrix used 56 further seeds, three demand files per seed and two controller arms, for 336 rollouts in total. Relative to the strict zero-source-row target model, the admitted controller reduced waiting by 4.527% (paired 95% interval, 4.279–4.787%); all 56 seed-level contrasts were favourable (two-sided Wilcoxon signed-rank probability value 7.5475×10^{-11}). The two frozen protocols are reported separately in Table 7.

The preselected controller outperformed its declared strict zero-source-row target comparator in Jinan, although that comparator has lower capacity. V98 therefore does not isolate source-row contribution at fixed architecture. Both arms also share the risk multiplier of 0.5 inherited from earlier Jinan closed-loop development. The result therefore does not establish a general multicity selector or superiority over PhasePressure.

A.7 Seven-city source-gate development

Three later source gates were frozen and evaluated on the seven development city groups. The robust-history gate selected a source for six targets, and the action-agreement gate selected a source for all seven; both improved the equal-city mean but failed the target-only safety comparison because their one-sided upper bounds were positive and their largest city regressions exceeded 0.02 normalized regret. A stricter mechanism-compatibility gate divided 25 target adaptation action groups into five folds. Each fit used 20 groups (B20), and the disjoint evaluation fold used five groups (B5), to score queue propagation, served movement, red accumulation, spillback and mobility. It admitted no source and returned exact target only in all seven cities (Table 11).

Table 11: Frozen seven-city source-gate development. Negative effects favour the selected gate. "Upper" is the one-sided 95% upper bound versus the architecture-matched target-only causal arm. Dense denotes the pooled H2O$^+$-style residual. These are development results, not external confirmation.

Version	Source gate	Source cities selected	Target comparator mean difference	Target upper 95% bound	Target cities improved	Maximum city regression	Dense comparator mean difference
V145	Robust history	6/7	-0.0028	+0.0093	4/7	+0.0281	-0.0720
V146	OOF action agreement	7/7	-0.0016	+0.0095	3/7	+0.0262	-0.0708
V148	Mechanism compatibility	0/7	+0.0000	+0.0000	0/7	+0.0000	-0.0691

The progression shows a trade-off between safety and usefulness. V145 and V146 retained source

signal but allowed harmful city tails. V148 removed those tails only by abstaining everywhere. Its 7/7 advantage over the dense residual is therefore inherited from the rigid target-only causal model and cannot be attributed to source-city labels or to a full mechanism world model. No external confirmation of these V145–V148 seven-city gate rules was authorized after their failures.

A.8 Post-V148 source-remedy diagnostics

We subsequently tested alternative explanations for this result without weakening the frozen safety or source-identity criteria (Table 12). Leave-one-seed diagnostics confirmed that beneficial source headroom exists after the fact. Mechanism-block priors, budget-aware shrinkage, local right-of-way context, state-conditioned utility, leave-city-out meta-utility and hierarchical priors nevertheless failed to turn that headroom into a stable deployment rule. Most conservative variants returned exact rigid fallback. The one state-conditioned gate that passed its offline development screen caused a 37.9% Cologne regression in the pre-expansion closed-loop smoke, so its full matrix was not launched. The later V123/V157A budget contrasts were also retained only as historical model-path comparisons after a Jinan domain-handling difference was found. V157B corrected B100 alone and passed its relative source-versus-target gate on the reused selector, while remaining worse than PhasePressure. The authorized V157C native one-action diagnostic completed seeds 80314 and 88625, but retained seed 27178 as invalid because the fixed 480-s checkpoint contained no eligible, scorable traffic light. Its two-valid-seed descriptive means were +0.00236 for source minus target, −0.00232 for source minus matched placebo, +0.00125 for target minus PhasePressure and +0.00362 for source minus PhasePressure. Only one of two seeds improved versus target and neither improved versus the PhasePressure heuristic; all recorded collision counts were zero. The fixed three-seed protocol is incomplete and supports no three-seed inference. V158 then froze the V157B predictors and collected 100 native-prefix groups (800 candidate rows and 700 non-reference branches) from ten fixed Jinan development seeds, with no simulator-state restoration. Five whole-seed folds gave source-minus-target −0.00113 (paired 95% interval $[-0.00240, +0.00019]$; 6/10 seeds improved), source-minus-placebo −0.00072 ($[-0.00219, +0.00091]$; 7/10) and source-minus-PhasePressure +0.00014 ($[-0.00239, +0.00261]$; 5/10). Every preregistered OOF comparison failed; the reserve was therefore neither authorized nor run. The failure closes this Jinan native-prefix ranking-remediation route and supports no controller adoption, unseen-city claim or post-hoc retuning.

V159 reused the exact V158 B100 bank and whole-seed folds, adding direct physical action endpoints without collecting another simulator trajectory. Endpoint target minus delta target was −0.00060648 (paired 95% interval, $[-0.00223628, +0.00081223]$), with 5/10 seeds improving. The corresponding differences in the base, `_2000` and `_2500` demand settings were −0.00201119, +0.00157150 and −0.00091152, respectively. Endpoint target remained +0.00065478 above PhasePressure, with only 3/10 seeds improving. The frozen gate failed; the direct-endpoint hypothesis is therefore closed without further retuning or a reserve run.

A.9 Metric population and safety accounting

SUMO `tripinfo` was written with unfinished records at the fixed 3,600-s horizon. The primary waiting-time population includes all departed vehicles, with unfinished waiting accrued to the horizon. Completed-only waiting and completion ratios are secondary outcomes. System vehicle-hours integrate active plus pending vehicles and therefore include demand not yet inserted. Across the complete matrix, 171 raw collision events collapsed to 13 unique participant–lane–time incidents in six rollouts; CFCMT had zero. All starting and ending teleport counters were zero.

Table 12: Post-V148 source-remedy ladder. Effects are normalized action-regret differences unless seconds or percentages are shown; negative values favour the source-aware arm. The V157C row reports descriptive means over two valid seeds because the fixed three-seed protocol was incomplete. The V158 row reports the preregistered ten-seed OOF estimates; its gate failed and the reserve was withheld. The V159 row reports the target-only direct-endpoint ablation; its OOF gate also failed. The versioned evidence identifiers are retained in the machine-readable source data, not in the displayed table.

Stage	Deployment information	Source	Held-out result	Adjudication
Repeatability diagnostic	Other development seeds; forced source	Oracle only	16/21 seed units and 7/7 city means improved	Headroom exists; not deployable
Mechanism parameter prior	Target folds plus source mechanism block	7/7 prototype	6/7 improved; mean -0.00409; placebo upper 95% +0.00089	Offline pass; source identity unresolved
Budget-aware shrinkage	B100 target folds; inverse-budget prior	1/7	mean -0.00523; no city regression	Reject multicity generality
Local mechanism context	B100 plus local right-of-way context	0/7	Exact rigid fallback in all cities	Reject
State-conditioned utility	B100 nested state-level utility labels	3/7	pre-correction historical: offline mean -0.00052; smoke: Cologne +37.9%, New York +2.6%, RESCO -17.7%	Historical; reject after closed-loop smoke
Feature-aligned dense utility	V150O replay; stored-name rigid binding	1/7 inner; 0 follow-up	macro +0.00135 vs rigid; Hangzhou +0.00942/+0.00061/+0.00093 vs rigid/placebo/blind	Reject globally and city-specifically
Leave-city-out meta utility	Six pseudo-target cities; target B100	0/7	Exact rigid fallback in all cities	Reject
Source-only hierarchical prior	Equal-city source prior; no target gate labels	0/7	forced mean +0.00277; 3/7 improved	Reject
Target-calibrated hierarchical prior	B100 target selector; same prior grid	0/7	forced mean -0.01213, 95% [-0.03879, +0.00266]	Reject; exact rigid fallback
Domain-aligned B100 source gate	V123/V157A confounded; reused 22-seed Jinan selector	One-action branch only	source-target -0.01357, 95% [-0.01784, -0.00935]; 20/22 improve; +0.02534 vs PhasePressure	B100 gate pass; V157C incomplete
Native one-action branch	Fixed Jinan roster; seed 27178 checkpoint at 480 s invalid	2/3 valid seeds	descriptive source-target +0.00236 (1/2 improve); source-placebo -0.00232; target-PhasePressure +0.00125; source-PhasePressure +0.00362 (0/2 improve); collisions 0; PhasePressure is the reference heuristic	Incomplete; no three-seed inference
Native-prefix ranking calibration	Frozen V157B predictors; native labels; five seed-held-out folds	OOF 0/3; no reserve	source-target -0.00113, 95% [-0.00240, +0.00019] (6/10 improve); source-placebo -0.00072, 95% [-0.00219, +0.00091] (7/10 improve); source-PhasePressure +0.00014, 95% [-0.00239, +0.00261] (5/10 improve); 100 groups/800 rows/700 branches; 0 restores	Reject; OOF gate failed; reserve withheld
Native endpoint ablation	V158 B100 target labels; direct physical endpoints	Target only; no source	endpoint-delta -0.00061, 95% [-0.00224, +0.00081] (5/10 improve); base/2000/2500 -0.00201/+0.00157/-0.00091; endpoint-PhasePressure +0.00065 (3/10 improve)	Reject; direct-endpoint hypothesis closed

A.10 Discarded pressure-guard development track

Before the anchored pairwise method was developed, an exploratory controller placed a learned mechanism residual behind a MaxPressure fallback. On eight RESCO scenarios and 600-s rollouts its mean queue was 12.918, close to MaxPressure at 12.907. At 3,600 s, PhasePressure was materially stronger (32.234 versus 43.511). These experiments motivated treating pressure as a hard performance ceiling, but they use a different controller, target set and metric from V43 and are excluded from the main confirmation.

Table 13: Exploratory 600-s pressure-guard stress test retained for development transparency. This is not the final anchored CFCMT controller.

Policy	Mean queue	Seed standard deviation	90th-percentile queue	Trip time	Trip wait	Time loss	Throughput
MaxPressure	12.9068	0.1546	19.5279	87.5891	8.0605	24.5171	0.8267
CFCMT + pressure guard	12.9180	0.1919	19.5973	87.4863	8.0114	24.4127	0.8256
PhasePressure	13.2446	0.1816	19.3911	86.5371	7.4544	23.4628	0.8199
H2O+-style dense	17.2279	3.6084	28.9325	86.6585	8.5768	24.5488	0.7653
Target-static simulator	20.4023	0.1758	32.5538	97.9633	19.9032	35.3928	0.7429
Native actuated SUMO	30.1825	0.3536	47.6873	119.7813	33.0017	57.0805	0.7580
Fixed RESCO program	42.6732	0.2709	69.6852	142.3116	54.6333	79.8759	0.7127

Table 14: Exploratory 3,600-s pressure-guard boundary check. PhasePressure is the strongest policy in this discarded protocol.

Policy	Mean queue	Seed standard deviation	90th-percentile queue	Trip time	Trip wait	Time loss	Throughput
PhasePressure	32.2343	0.9901	62.1800	104.2291	20.8528	38.5434	0.9492
MaxPressure	42.7872	0.7314	71.6277	106.9182	20.5521	40.7331	0.9448
CFCMT + pressure guard	43.5106	0.5699	74.2496	108.6326	22.0400	42.4485	0.9448
Target-static simulator	44.8973	1.2335	65.4094	258.2547	169.8900	188.9974	0.8754
H2O+-style dense	49.5058	3.9510	74.9880	221.4173	131.1706	150.3275	0.8827
Native actuated SUMO	58.3468	1.1370	99.4055	164.5754	66.1326	98.1604	0.9714
Fixed RESCO program	84.6088	0.6458	133.0312	221.2093	113.3988	154.3776	0.9385

A.11 Post-freeze fresh-seed execution-layer stress test

The V86–V89 successor track uses a target-state-conditioned 120-s action originator and global execution cooldown rather than the V43 anchored all-pair controller. Negative deltas favour the learned executor relative to PhasePressure. The first three rows reuse the same 54 disclosed adaptive seeds and are not independent replications. Only V89 uses 64 newly generated seeds; its joint gate was frozen before those rollouts and failed. The eight V85 prospective seeds were not opened.

Table 15: Post-freeze Jinan execution-layer stress test. Intervals are paired simulator-seed bootstrap intervals. Development rows are shown to expose the adaptive-to-confirmatory shrinkage and are not pooled with V89.

Stage	Evidence	Seeds	Mean relative waiting difference	95% interval	Improved	Decision
V86 legacy global	adaptive screen	54	-0.429%	[-0.818, -0.033]%	68.5%	development only
V87 unbounded stay	adaptive ablation	54	0.104%	[-0.295, 0.519]%	46.3%	variant rejected
V88 one-stay budget	adaptive ablation	54	-0.242%	[-0.674, 0.195]%	55.6%	variant rejected
V89 frozen legacy	fresh confirmation	64	-0.043%	[-0.449, 0.370]%	53.1%	confirmation rejected

Table 16: Official RESCO full-budget RL results under the upstream native protocol. Unsupported upstream scenario–algorithm pairs are not silently patched into new methods. These rows are not same-protocol comparisons with CFCMT.

Algorithm	Scenario	Runs	Time loss	Duration	Waiting	Elapsed (s)
IDQN	arterial4x4	3	1505.536	501.305	436.920	17179.205
MPLight	arterial4x4	3	1056.797	357.454	238.745	13576.024
IDQN	cologne1	3	612.737	164.905	138.950	812.441
MPLight	cologne1	3	189.393	122.751	68.843	6985.784
IDQN	cologne3	3	86.610	92.313	43.268	2351.500
MPLight	cologne3	3	1011.239	543.511	515.329	10132.753
IDQN	cologne8	3	24.760	89.379	8.354	5946.643
IDQN	grid4x4	3	47.752	158.215	25.205	11120.623
MPLight	grid4x4	3	44.393	156.112	16.598	10392.985
IDQN	ingolstadt1	3	18.097	37.191	5.490	911.332
MPLight	ingolstadt1	3	401.895	85.906	63.192	7888.143
IDQN	ingolstadt21	3	256.588	353.111	176.405	13141.256
IDQN	ingolstadt7	3	51.869	78.827	17.357	5313.646
MPLight	ingolstadt7	3	284.391	111.560	57.463	8755.139

Table 17: Patched LibSignal full-budget execution audit under native LibSignal states, rewards, timing and metrics.

Agent	Network	Runs	Travel time	Queue	Delay	Throughput	Elapsed (s)
dqn	sumo1x1	3	38.945	3.240	0.283	1998.667	175.967
frap	sumo1x1	3	41.923	24.461	0.460	1664.667	439.620
mplight	sumo1x1	3	40.546	4.373	0.319	1995.000	371.569
presslight	sumo1x1	3	39.748	3.442	0.285	1998.667	175.835
dqn	sumo1x3	3	97.090	6.531	0.244	2487.333	425.384
frap	sumo1x3	3	118.712	11.672	0.398	2505.333	1583.053
mplight	sumo1x3	3	66.502	1.876	0.243	2814.333	647.664
presslight	sumo1x3	3	63.188	1.164	0.188	2816.000	393.414
dqn	sumo4x4	3	140.610	0.603	0.041	1465.000	2140.773
frap	sumo4x4	3	140.477	0.612	0.042	1464.000	7188.185
mplight	sumo4x4	3	140.469	0.607	0.042	1464.333	1017.297
presslight	sumo4x4	3	144.279	0.711	0.048	1463.333	2080.043

A.12 Native-protocol RL checks

A.13 Frozen artifact chain

The publication artifacts are regenerated with

```
python3 paper_tsc/figures/build_tsc_external_confirmation.py
python3 paper_tsc/figures/build_tsc_causal_transfer_schematic.py
python3 paper_tsc/figures/build_tsc_postfreeze_stress_test.py
python3 paper_tsc/figures/build_tsc_source_gate_development.py
python3 paper_tsc/figures/build_tsc_source_remedy_ladder.py
```

Table 18: SHA-256 chain for the final external result. The paper artifact builder verifies all result gates before regenerating this table.

Artifact	SHA-256
development audit	5bcae9467cdb4cc157676e94078d204702ebdca6d51007ffca4bc85d13951066
source cache	14c6dfad87999e55fa86fa942ac0ca2334129f4d1662b3c31a6e4fc7aa74c242
target adaptation cache	beb6a968c43dd58ceac4df6d9da8dfca47fc1f0869dfda0dc151a3e9cae58d77
joint method freeze	471e8c34eb7be96411c812c7fe961d5c07f1857e3fb50d2c8d7edd59fc23c596
held-out seed-8171 result	20d682b48bca0f8821ecf98e54b04c94af87d530add51cc1d92d9a176f74cf60
closed-loop aggregate	be36c73c5835e632231140fdbdabc4bba422f54b503160c0179efc902d4534e0
post-freeze target-only ablation	238dbdc27ceb7ce016b9c995b5472cf1ad71a76632b6609b0df66408a1e26a7e
V91 legacy near-target-only confirmation	3a0ee874afeff6df96a90fb4aed4e42f79705469f0d2a72da514932385990f67
V98 preselected controller-pair confirmation	bb00da6d3ee0f9420bdd7827eeada7c0b4755fadae4a871e3de6c42ded1dc65d
V98 remote result identity audit	f8243b8279e600721f978a763415ced656c3528b190fb952877b17f2b982a91b
Los Angeles deployment model	f26ee8e5829e248ef411f77ad3e19ed0d63fd52cdfdae6d8291a9a3aeaaa7aff
Jinan deployment model	a34970f1e4f023079650866097359ec3b2509e17787685249a21363cc7b03741
closed-loop source tree	ad555ff72a11a62cc88c8bbbe2ab2f5df1db8de920af03e2f61d99aaba226a7d

```
python3 paper_tsc/figures/build_v43_dense_fresh10_confirmation.py
```

These commands generate the submission figures and source tables. The unreferenced V16-era builder and its outputs remain development history; they are not used for the submission figures in Figures 1 to 5. The first command refuses to write figures or tables if the development, external offline, joint-freeze or closed-loop validity gate is not satisfied. The third command checks all V86–V89 evidence-integrity audits and preserves the preregistered V89 rejection. The fourth command preserves the V145, V146 and V148 development rejections. The fifth command checks that the canonical post-V148 remedy artifacts retain their recorded cohorts and adjudications, including the failed closed-loop smoke, exact rigid fallbacks, the superseded V123/V157A comparison, the bounded V157B B100 branch authorization and the incomplete V157C three-seed native one-action diagnostic. It also requires the V158 native-prefix OOF gate and the V159 direct-endpoint OOF gate to remain failed, with both reserve authorizations false. The sixth command independently regenerates the frozen-model ten-new-seed figure and its source tables from the registered protocol and complete 80-rollout report, without changing the original confirmation artifacts.

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